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M3dmeas functions

Synopsis

The functions prefixed with M3dmeas make up the Aurora Imaging Library 3D Measurement module.

Functions

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M3dmeasAlloc

Synopsis

Allocate a 3D measurement context.

Syntax

```
AIL_ID M3dmeasAlloc(
    AIL_ID SysId,              //in
    AIL_INT64 ContextType,     //in
    AIL_INT64 ControlFlag,     //in
    AIL_ID *Context3dmeasIdPtr //out
)
```

Description

This function allocates a 3D measurement context on the specified system. A 3D measurement context contains information needed to perform an M3dmeasFindMarker() or M3dmeasFit() operation, which allow you to find markers in a specified depth map or fit a geometry to found markers, respectively. Markers denote the location of edge transitions along a profile; the edge transition can go from a lower Z-value to a higher Z-value (or vice versa). You can also use **M3dmeasAlloc**() to allocate a draw 3D measurement context for drawing results using M3dmeasDraw3d().

You can find markers in a profile taken along a specified path, in profiles taken at regular intervals perpendicular to a specified template, or in each row of a depth map, where each row is interpreted as a single profile.

Finding markers along a path is useful when you want to find an object but you don't know its exact location. In this case, you can allocate an M_FIND_MARKER_PATH_CONTEXT and use M3dmeasDefine() to set a path that crosses through the possible locations. The 3D measurement module will extract a profile along the path and find the markers within the profile.

Finding markers perpendicular to a template is useful when you want to verify the location or shape of an object. You can allocate an M_FIND_MARKER_TEMPLATE_CONTEXT and specify a template that follows an expected edge of your object using M3dmeasDefine(). To find the markers that can form the template, the 3D measurement module will extract profiles at regular intervals perpendicular to the template and find a single marker in each profile where the height change occurs. Note that you can use M3dmeasFit() to fit a geometry to the found markers to compare their location to that of the template.

You can use a profile 3D measurement context (M_FIND_MARKER_PROFILE_CONTEXT) to analyze an entire depth map. In this case, the profile extraction step is skipped and the find marker operation is performed on each row (profile) of the depth map. The 3D measurement module can find the specified number of markers along each profile. Note that a profile 3D measurement context contains a default template that is inherent to the supplied profiles and does not need to be defined.

After allocating the 3D measurement context, and if required, defining the path or template, use M3dmeasFindMarker() or M3dmeasFit() to find and/or fit the markers.

When the 3D measurement context is no longer required, release it using M3dmeasFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the 3D measurement context's lifetime and you must not manually free it.

Parameters

SysId

Specifies the system on which to allocate the 3D measurement context.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ContextType

Specifies the type of 3D measurement context to allocate. This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the context object																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_3D_PATH_CONTEXT	Specifies to allocate a draw path 3D measurement context for use with M3dmeasDraw3d(), to draw features of a path, a path's profile, or markers found in a path's profile.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_3D_PROFILE_CONTEXT	Specifies to allocate a draw profile 3D measurement context for use with M3dmeasDraw3d(), to draw features of supplied profiles or markers found in supplied profiles.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_3D_TEMPLATE_CONTEXT	Specifies to allocate a draw template 3D measurement context for use with M3dmeasDraw3d(), to draw features of a template, a template's perpendicular profiles, markers found in a template's perpendicular profiles, or results from a fit operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_MARKER_PATH_CONTEXT	Specifies to allocate a path 3D measurement context for use with M3dmeasFindMarker(), to find markers in a path's profile.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_MARKER_PROFILE_CONTEXT	Specifies to allocate a profile 3D measurement context for use with M3dmeasFindMarker(), to find markers in supplied profiles. Note that supplied profiles have a corresponding default template that does not need to be defined.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_MARKER_TEMPLATE_CONTEXT	Specifies to allocate a template 3D measurement context for use with M3dmeasFindMarker() or M3dmeasFit(), to find markers in a template's perpendicular profiles.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_CONTEXT	Specifies to allocate a fit 3D measurement context for use with M3dmeasFit(), to fit a geometry to markers found in a template's perpendicular profiles.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Context3dmeasIdPtr

Specifies the address of the variable in which to write the 3D measurement context identifier or specifies the data type that the function should use to return the 3D measurement context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D measurement context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D measurement context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DMEAS_ID* is returned instead of a standard Aurora Imaging Library identifier. This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D measurement context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the draw path 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated draw path 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the draw profile 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated draw profile 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the draw template 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated draw template 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the fit 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated fit 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Address in which to write the path 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated path 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the profile 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated profile 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the template 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated template 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 3D measurement context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DMEAS_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasAllocResult

Synopsis

Allocates a 3D measurement result buffer.

Syntax

```
AIL_ID M3dmeasAllocResult(
    AIL_ID SysId,             //in
    AIL_INT64 ResultType,     //in
    AIL_INT64 ControlFlag,    //in
    AIL_ID *Result3dmeasIdPtr //out
)
```

Description

This function allocates a 3D measurement result buffer on the specified system, to store results obtained from an M3dmeasFindMarker() or M3dmeasFit() operation.

When the 3D measurement result buffer is no longer required, release it using M3dmeasFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the 3D measurement result buffer's lifetime and you must not manually free it.

Parameters

SysId

Specifies the identifier of the system on which to allocate the 3D measurement result buffer.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of 3D measurement result buffer to allocate.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of result to allocate																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FIND_MARKER_PATH_RESULT	Specifies to allocate a path 3D measurement result buffer, used to store M3dmeasFindMarker() results.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIND_MARKER_PROFILE_RESULT	Specifies to allocate a profile 3D measurement result buffer, used to store M3dmeasFindMarker() results.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIND_MARKER_TEMPLATE_RESULT	Specifies to allocate a template 3D measurement result buffer, used to store M3dmeasFindMarker() or M3dmeasFit() results.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Result3dmeasIdPtr

Specifies the address of the variable in which to write the 3D measurement result buffer identifier or specifies the data type that the function should use to return the 3D measurement result buffer identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D measurement result buffer ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D measurement result buffer ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DMEAS_ID* is returned instead of a standard Aurora Imaging Library identifier. This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D measurement result buffer (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the path 3D measurement result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated path 3D measurement result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the profile 3D measurement result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated profile 3D measurement result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the template 3D measurement result buffer identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated template 3D measurement result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 3D measurement result buffer identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DMEAS_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasControl

Synopsis

Control a setting for a 3D measurement context or result buffer.

Syntax

```
void M3dmeasControl(
    AIL_ID ContextOrResult3dmeasId, //out
    AIL_INT64 PathOrTemplateIndex,  //in
    AIL_INT64 ProfileIndex,         //in
    AIL_INT64 ControlType,          //in
    AIL_DOUBLE ControlValue         //in
)
```

Description

This function allows you to control a setting of a 3D measurement context or result buffer. You can typically inquire these settings using M3dmeasInquire().

To control draw 3D measurement settings for M3dmeasDraw3d(), use M3dmeasControlDraw() instead.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **M3dmeasControl()** internally calls the *AIL_DOUBLE* version of this function (M3dmeasControlDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (M3dmeasControlInt64()).

Parameters

ContextOrResult3dmeasId

Specifies the identifier of the 3D measurement context or result buffer to control. The 3D measurement context or result buffer must have been previously allocated on the required system using M3dmeasAlloc() or M3dmeasAllocResult().

PathOrTemplateIndex

Specifies to control a 3D measurement context, a path, a template, a default template, or a 3D measurement result buffer. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying what to control																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default value. If a 3D measurement context is specified, same as M_CONTEXT. If a 3D measurement result buffer is specified, same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_PATH_INDEX(AIL_INT PathIndex)	Specifies to control an individual path in a path 3D measurement context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PathIndex Set this parameter to the following:															

	0 Specifies the index of the path.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies to control an individual template in a template 3D measurement context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	TemplateIndex Set this parameter to the following:															
	0 Specifies the index of the template.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT	Specifies to control a global setting of a 3D measurement context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_TEMPLATE	Specifies to control the default template in a profile 3D measurement context, if one is specified. Note that the default template in a profile 3D measurement context is different from an explicitly defined template in a template 3D measurement context. The default template is inherent to the supplied profiles, such that it is the theoretical template that would result in these profiles. Note, unlike for the profiles perpendicular to the template in a template 3D measurement context, multiple markers can be found along the profiles perpendicular to the default template in a profile 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to control a general setting of a 3D measurement result buffer, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b

ProfileIndex

Specifies the profile to control, if required. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a profile																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the index of a profile is not required.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_INDEX(AIL_INT ProfileIndex)	Specifies to control an individual profile in a template 3D measurement context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	ProfileIndex Set this parameter to one of the following:															

			M_ALL	Specifies all profiles.	c		e	f	g	a		i		j	k	l	m	n	b
			Value >= 0	Specifies the index of the profile.	c		e	f	g	a		i		j	k	l	m	n	b

ControlType

Specifies the setting to control.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the setting's new value.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **ControlType** and **ControlValue** parameters.

- To control a global setting of a profile, path, or template 3D measurement context
- To control the default template in a profile 3D measurement context, path in a path 3D measurement context, or template in a template 3D measurement context
- To control the profile extraction along a path in a path 3D measurement context or perpendicular to the template in a template 3D measurement context
- To control a template in a template 3D measurement context
- To control a template or profile in a template 3D measurement context
- To control a profile in a template 3D measurement context
- To control a fit 3D measurement context
- To control a profile, path, or template 3D measurement result buffer

The following ControlType and ControlValue parameter setting can be specified to control a global setting of a profile, path, or template 3D measurement context. PathOrTemplateIndex must be set to M_DEFAULT or M_CONTEXT, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To control a global setting of a profile, path, or template 3D measurement context																		
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																	
M_TIMEOUT		Sets the maximum amount of time for M3dmeasFindMarker() or M3dmeasFit() to complete the find marker operation before generating a time-out error. INQ																
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and ControlValue parameter settings can be specified for the default template in a profile 3D measurement context, a path in a path 3D measurement context, or a template in a template 3D measurement context. PathOrTemplateIndex must be set to M_DEFAULT_TEMPLATE, the index of a path, or the index of a template, respectively, and

ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

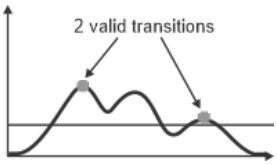
To control the default template in a profile 3D measurement context, path in a path 3D measurement context, or template in a template 3D measurement context

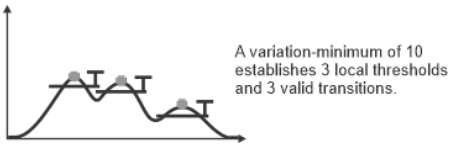
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_MARKER_DEPTH_MIN		Sets the minimum depth required for a peak or valley in the profile response, for it to be considered a valid transition signifying a marker. The edge profile response is established by taking the normalized first derivative of the profile, which generally resembles a peak when a valid edge transition is present. The depth of an edge transition is measured as the difference between the Z-coordinates at the start and end of the transition. The ridge profile response is established by taking the normalized second derivative of the profile, which generally resembles a peak or valley when a valid ridge transition is present. The depth of a ridge transition is measured as the difference between the Z-coordinate at the summit of the peak/bottom of the valley and the closest Z-coordinate at the start or end of the transition. INQ															
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the minimum depth. A setting of 0.0 indicates that Aurora Imaging Library considers all depths.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_NUMBER		Sets the number of markers to search for in the profile. Note that this control type is not available for a template 3D measurement context. In this case, the find marker operation always looks for 1 marker in each perpendicular profile of the template. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies to search for all markers.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of markers for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_NUMBER_TRANSITION_INSIDE		Sets the number of transitions to find inside a marker. Note that border transitions are not included. This control type is only available when M_MARKER_TYPE is set to M_PAIR. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies to find markers with any number of transitions.	c		e	f	g	a		i		j	k	l	m	n	b

	Value > 0	Specifies the number of transitions.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_POLARITY		Sets the polarity of the transition to find. This is the polarity of the sole transition for an M_SINGLE marker and the first transition for an M_PAIR marker. Transitions from a lower Z-value to a higher Z-value, according to the direction of the search along the profile, have a positive polarity. Transitions from a higher Z-value to a lower one have a negative polarity. If the polarity does not matter, you can set this control to M_ANY to find transitions with either a positive or negative polarity. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_POSITIVE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies to find a transition with any polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NEGATIVE	Specifies to find a transition with negative polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITIVE	Specifies to find a transition with positive polarity.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_POLARITY_SECOND		Sets the polarity of the second transition to find. This is the polarity of the second transition for an M_PAIR marker. Transitions from a lower Z-value to a higher Z-value, according to the direction of the search along the profile, have a positive polarity. Transitions from a higher Z-value to a lower one have a negative polarity. If the polarity does not matter, you can set this control to M_ANY to find transitions with either a positive or negative polarity. If you want to find transitions with the same or opposite polarity as M_MARKER_POLARITY, you can set this control to M_SAME or M_OPPOSITE, respectively. This control type is only available when M_MARKER_TYPE is set to M_PAIR. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_OPPOSITE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies to find a transition with any polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NEGATIVE	Specifies to find a transition with negative polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPPOSITE	Specifies to find a transition with the opposite polarity of the first transition.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITIVE	Specifies to find a transition with positive polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies to find a transition with the same polarity as the first transition.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_POSITION_INVALID		Sets where to identify a transition when an invalid region is encountered in the profile. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_BOTH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AFTER	Specifies that the transition is immediately after the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BEFORE	Specifies that the transition is immediately before the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOTH	Specifies that there is one transition immediately before and one transition immediately after the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIDDLE	Specifies that the transition is in the middle of the gap.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_POSITION_RELATIVE		Sets the position that the markers should be closest to when M_MARKER_SELECTION is set to M_POSITION_RELATIVE and the number of candidates is greater than M_MARKER_NUMBER. Note that this also sets the relative position to use when calculating the M3dmeasGetResult() M_OFFSET result type. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the position, as a percentage of the nominal profile length.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_SELECTION		Sets the policy used to select found markers when the number of candidates is greater than M_MARKER_NUMBER. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_STRONGEST.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BEST_FIT	Specifies to use a different selection criteria depending on the operation. When performing an M3dmeasFindMarker() operation, M_BEST_FIT specifies to select markers according to their proximity to the profile's origin; the closest marker is selected first. When performing an M3dmeasFit() operation with a template 3D measurement context, M_BEST_FIT specifies to select markers according to possible resulting fit scores; the marker that would result in the best fit possible is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSEST_TO_SPACING	Specifies to select markers according to their similarity to the spacing set using M_MARKER_SPACING_RELATIVE; the marker with the closest spacing is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSEST_TO_WIDTH	Specifies to select markers according to their similarity to the width set using M_MARKER_WIDTH_RELATIVE; the marker with the closest width is selected first.	c		e	f	g	a		i		j	k	l	m	n	b

	M_NARROWEST	Specifies to select markers according to their width; the narrowest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_FIRST	Specifies to select markers according to their proximity to the profile's origin; the closest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_LAST	Specifies to select markers according to their proximity to the end of the profile; the closest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_RELATIVE	Specifies to select markers according to their proximity to the position set using M_MARKER_POSITION_RELATIVE; the closest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STRONGEST	Specifies to select markers according to the strength of the transition; the marker with the strongest transition is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TALLEST	Specifies to select markers according to their measured depth; the marker with the greatest depth is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WIDEST	Specifies to select markers according to their width; the widest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_SPACING_RELATIVE		Sets the spacing that the marker's spacing should be closest to when M_MARKER_SELECTION is set to M_CLOSEST_TO_SPACING and the number of candidates is greater than M_MARKER_NUMBER. The spacing of a marker is the distance between the center of the previous marker and the center of the next marker. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the spacing, as a percentage of the nominal profile length.	c		e	f	g	a		i		j	k	l	m	n	b

M_MARKER_STRENGTH_MIN	Sets the minimum strength required for a peak or valley in the profile response, for it to be considered a valid transition signifying a marker. The edge profile response is established by taking the normalized first derivative of the profile, which generally resembles a peak when a valid edge transition is present. The ridge profile response is established by taking the normalized second derivative of the profile, which can resemble a peak or valley when a valid ridge transition is present. For peaks, the summit of the highest peak that rises above and then falls below the minimum strength on both sides of the peak represents the transition's position. For valleys, the transition's position is at the bottom of the lowest valley that falls below and then rises above the negative of the minimum strength on both sides of the valley. The strength is the absolute value at these positions.  When searching for multiple transitions, you might need to further constrain the strengths that Aurora Imaging Library considers, using M_MARKER_STRENGTH_MIN_VAR. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 4.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the minimum strength. A setting of 0.0 indicates that Aurora Imaging Library considers all strengths, while a setting of 100.0 indicates that Aurora Imaging Library only considers the greatest possible strength. The greatest possible strength results from a perfectly straight transition from the lowest Z-value to the highest Z-value (or vice versa) over one pixel in the profile response image. If you set the minimum strength to a very low value, M3dmeasFindMarker() can take longer to execute.	c		e	f	g	a		i		j	k	l	m	n	b

M_MARKER_STRENGTH_MIN_VAR		Sets the minimum prominence required for a peak or valley in the profile response, for it to be considered a valid transition signifying a marker. This control is useful for advanced applications where you are trying to find multiple neighboring transitions in the profile, and M_MARKER_STRENGTH_MIN does not give you enough control over identifying the required transitions. To determine if the minimum prominence is met, Aurora Imaging Library establishes a local threshold for each peak or valley in the profile response. A peak's threshold is its value minus M_MARKER_STRENGTH_MIN_VAR. A valley's threshold is its value plus M_MARKER_STRENGTH_MIN_VAR.  The portion of the profile response above the local threshold that contains the peak, or below the local threshold that contains the valley, can be referred to as the interval of the peak/valley, provided that the profile response eventually returns to the other side of the local threshold on both sides. For Aurora Imaging Library to consider a peak or valley to be a valid transition, there must be no strength greater than it within its interval. For peaks, if the local threshold calculated with M_MARKER_STRENGTH_MIN_VAR is less than M_MARKER_STRENGTH_MIN, the local threshold equals M_MARKER_STRENGTH_MIN. For valleys, the local threshold equals the negative of M_MARKER_STRENGTH_MIN if it is greater. Note that this control type is most useful for edge transitions. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 100.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the minimum prominence. A setting of 0.0 indicates no prominence; that is, Aurora Imaging Library only considers the maximum strength of the peak/valley's interval. A setting of 100.0 (default) indicates the greatest possible prominence; that is, Aurora Imaging Library considers all strengths of the interval that exceed M_MARKER_STRENGTH_MIN. In this case, M_MARKER_STRENGTH_MIN_VAR has no effect.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_TRANSITION		Sets the type of transition to find. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_EDGE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGE	Specifies to find edge transitions.	c		e	f	g	a		i		j	k	l	m	n	b

	M_EDGE_OR_INVALID	Specifies to find edge transitions and invalid transitions. Use M_MARKER_POSITION_INVALID to specify whether invalid transitions are identified before, after, or in the middle of gaps encountered in the profile.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INVALID	Specifies to find invalid transitions. Use M_MARKER_POSITION_INVALID to specify whether invalid transitions are identified before, after, or in the middle of gaps encountered in the profile.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RIDGE	Specifies to find ridge transitions.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_TYPE		Sets the type of marker to extract from the profile. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SINGLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PAIR	Specifies to extract markers represented by a pair of transitions.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SINGLE	Specifies to extract markers represented by a single transition.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_WIDTH_RELATIVE		Sets the width that the marker's width should be closest to when M_MARKER_SELECTION is set to M_CLOSEST_TO_WIDTH and the number of candidates is greater than M_MARKER_NUMBER. The width of an M_SINGLE marker is the distance between the start and end of the transition. The width of an M_PAIR marker is the distance between the center of the first transition and center of the second transition. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the width, as a percentage of the nominal profile length.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_FILTER_SMOOTHNESS		Sets the normalized degree of smoothness (strength of denoising) applied to the profile to create the profile response before finding the markers. Increasing the smoothness value reduces transitions caused by noise, lowering the risk of finding false markers. It can also increase the strength of gradual transitions, allowing some previously undetected markers to be found. This control type is only used when M_PROFILE_FILTER_SMOOTHNESS_TYPE is set to M_NORMALIZED. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the smoothness value. A value of 100.0 results in a strong noise reduction effect, while a value of 0.0 has almost no noise reduction effect.	c		e	f	g	a		i		j	k	l	m	n	b

M_PROFILE_FILTER_SMOOTHNESS_SIZE		Sets the smoothness size applied to the profile to create the profile response before finding the markers. A size close to 3 times the transition's width (M_WIDTH * 3) is recommended, particularly for ridge transitions. If the size is too large, transition strengths will decrease. For edge transitions, increasing the smoothness size reduces transitions caused by noise, lowering the risk of finding false markers. It can also increase the strength of gradual transitions, allowing some previously undetected markers to be found. This control type is only used when M_PROFILE_FILTER_SMOOTHNESS_TYPE is set to M_SIZE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 3.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 2.0	Specifies the smoothness size.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_FILTER_SMOOTHNESS_TYPE		Sets the filter smoothness type. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_NORMALIZED.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORMALIZED	Specifies to apply a normalized filter smoothness, set with M_PROFILE_FILTER_SMOOTHNESS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SIZE	Specifies to apply a filter smoothness size, set with M_PROFILE_FILTER_SMOOTHNESS_SIZE.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_GAP_FILL_SHARP_ELEVATION		Sets how to fill sharp elevation gaps. A gap is considered a sharp elevation gap if the difference between the values on either side of the gap is at least M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH. You can specify to propagate the boundary value before or after the gap, propagate the minimum or maximum gap boundary value, propagate both boundary values up to the middle of the gap, or not to fill sharp elevation gaps. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AFTER	Specifies to fill the gap with the value of the boundary after the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BEFORE	Specifies to fill the gap with the value of the boundary before the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to fill sharp elevation gaps.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX	Specifies to fill the gap with the maximum gap boundary value.	c		e	f	g	a		i		j	k	l	m	n	b

	M_MIDDLE	Specifies to fill the first half with the value of the boundary before the gap and the second half with the value of the boundary after the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN	Specifies to fill the gap with the minimum gap boundary value.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH		Sets the threshold used to differentiate between gradual elevation gaps and sharp elevation gaps. A gap is an invalid region in the profile resulting from missing data in the source depth map. If the difference between a gap's boundary values is less than the specified threshold, the gap is considered a gradual elevation gap, otherwise it is considered a sharp elevation gap. Gradual elevation gaps are filled using linear interpolation; sharp elevation gaps are either filled by propagating the boundary values or are left unfilled, depending on the M_PROFILE_GAP_FILL_SHARP_ELEVATION control type. It is often better to fill gradual elevation gaps using linear interpolation since the underlying surface of such a gap is considered to be part of the same surface as its boundaries. Whereas for sharp elevation gaps, the underlying surface is considered to be discontinuous at least at one of its boundaries, so propagating the value of one of the boundaries is recommended. If a gap is greater than M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH, it is replaced using the secondary algorithm described in M_PROFILE_GAP_FILL_SHARP_ELEVATION. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies that the threshold is infinite. When M_INFINITE is specified, all gaps are considered gradual elevation gaps, so linear interpolation is always used and M_PROFILE_GAP_FILL_SHARP_ELEVATION is ignored.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the threshold, as a percentage of the depth map's Z-range or in world units, depending on the units specified with M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH_UNITS. When the threshold is set to 0.0, all gaps are considered sharp elevation gaps, so linear interpolation is never used and all gaps are filled according to M_PROFILE_GAP_FILL_SHARP_ELEVATION.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH_UNITS		Sets the units with which to interpret the M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH control type. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_PERCENTAGE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PERCENTAGE	Specifies to interpret the value as a percentage of the range of Z-values (Z-range) visible in the depth map.	c		e	f	g	a		i		j	k	l	m	n	b

	M_WORLD	Specifies to interpret the value in world units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_GAP_FILL_THRESHOLD		Sets the maximum size of gaps that will be filled. Gaps are invalid regions in the profile where the 3D sensor could not pick up information, due to occlusion, reflections, or areas out of the camera's field of view. If a gap in the profile is larger than the fill threshold, it is not replaced. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no maximum gap size.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the maximum gap size, in sample or world units, depending on the units specified with M_PROFILE_GAP_FILL_THRESHOLD_UNITS.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_GAP_FILL_THRESHOLD_UNITS		Sets the units with which to interpret the M_PROFILE_GAP_FILL_THRESHOLD control type. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SAMPLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAMPLE	Specifies to interpret the value relative to the sampling distance (M_PROFILE_SAMPLE_SIZE).	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD	Specifies to interpret the value in world units.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and ControlValue parameter settings can be specified for a path in a path 3D measurement context or a template in a template 3D measurement context. PathOrTemplateIndex must be set to the index of a path or template, respectively, and ProfileIndex must be set to M_DEFAULT. These control types are used to specify how to extract the profile along the path or the profiles at regular intervals perpendicular to the template. Note that the sampling distance determines where each profile point will be extracted. The points within the thickness of the profile line at each of these positions are considered a sample, and a single profile point is calculated from the points in each sample.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To control the profile extraction along a path in a path 3D measurement context or perpendicular to the template in a template 3D measurement context																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_PROFILE_ACCUMULATE_TYPE		Sets how points in a given sample are accumulated to create a single profile point when there are multiple points within the thickness of the profile line. INQ															
	M_DEFAULT	Same as M_MEAN.	c		e	f	g	a		i		j	k	l	m	n	b

	M_MAX	Specifies to use the largest value of all the points in the sample. > This option is not supported when M_PROFILE_INTERPOLATION_MODE is set to M_AVERAGE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEAN	Specifies to use the mean value of all the points in the sample.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN	Specifies to use the smallest value of all the points in the sample. > This option is not supported when M_PROFILE_INTERPOLATION_MODE is set to M_AVERAGE.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_INTERPOLATION_AVERAGE_FRACTION		Sets the fraction of the sampling distance to use when taking a weighted average of the area surrounding the profile point. For example, a value of 0.5 indicates that 50 percent of the pixel area that surrounds the profile point (defined by 50% of the sampling distance along the line) is used to calculate a weighted average for the profile point. This control type is only available when M_PROFILE_INTERPOLATION_MODE is set to M_AVERAGE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 1.0	Specifies the fraction.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_INTERPOLATION_MODE		Sets the interpolation mode used to calculate the value for each point in the sample that will be used to create the profile point. Note that when averaging interpolation is used (M_AVERAGE), the profile point is calculated without calculating the value for each point in the sample. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_BILINEAR.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AVERAGE	Specifies averaging interpolation. The value of the profile point is determined by taking a weighted average of the pixel area that surrounds the profile point. You can use M_PROFILE_INTERPOLATION_AVERAGE_FRACTION to specify how much of the pixel area to consider.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BILINEAR	Specifies bilinear interpolation. The new value is determined by taking a weighted average of the 4 values (2x2) that surround the point in the sample.	c		e	f	g	a		i		j	k	l	m	n	b

	M_NEAREST_NEIGHBOR	Specifies nearest neighbor interpolation. The new value is that of the pixel closest to the point in the sample.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_MIN_VALID_PERCENTAGE		Sets the minimum percentage of points in a given sample that must be valid to set the corresponding profile point to a valid value. When there are multiple points within the thickness for the sample that will be used to create the profile point, the profile point will be invalid if the minimum valid percentage is not met. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0%.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the percentage of points in the sample. Note that a value of 0.0% means there are no constraints on the number of valid points, but if all points contributing to the calculation of a profile point are invalid, the profile point will be invalid in the destination.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE		Sets the nominal angle at which the projection is done for a given profile. This is the angle, relative to the profile line, from which samples are taken for a profile point. The projection angle determines the angle of the markers that are found. You should set this control type to the angle at which you expect to find the markers, relative to the profile line. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 90.0°.	c		e	f	g	a		i		j	k	l	m	n	b
	10.0 <= *Value* <= 170.0	Specifies the nominal projection angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE_ACCURACY		Sets the accuracy of the projection angle. The accuracy defines an angular range in which to fine-tune the projection after the approximate angle with the best markers is found. To be effective, you must set the degree of accuracy to a value smaller than that of the rotation tolerance. When M_PROFILE_PROJECTION_ANGLE_MODE is disabled, this setting has no effect.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0°.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 170.0	Specifies the accuracy, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_PROFILE_PROJECTION_ANGLE_DELTA_NEG	Sets the lower limit of the projection angle range, relative to the nominal projection angle (M_PROFILE_PROJECTION_ANGLE). That is, *LowerLimit* = M_PROFILE_PROJECTION_ANGLE - M_PROFILE_PROJECTION_ANGLE_DELTA_NEG. Note that the minimum projection angle considered is 10.0°, even if M_PROFILE_PROJECTION_ANGLE - M_PROFILE_PROJECTION_ANGLE_DELTA_NEG is less than 10.0°. When M_PROFILE_PROJECTION_ANGLE_MODE is disabled, this setting has no effect. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 5.0°.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 170.0	Specifies the lower limit of the projection angle range, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE_DELTA_POS	Sets the upper limit of the projection angle range, relative to the nominal projection angle (M_PROFILE_PROJECTION_ANGLE). That is, *UpperLimit* = M_PROFILE_PROJECTION_ANGLE + M_PROFILE_PROJECTION_ANGLE_DELTA_POS. Note that the maximum projection angle considered is 170.0°, even if M_PROFILE_PROJECTION_ANGLE + M_PROFILE_PROJECTION_ANGLE_DELTA_POS is greater than 170.0°. When M_PROFILE_PROJECTION_ANGLE_MODE is disabled, this setting has no effect. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 5.0°.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 170.0	Specifies the upper limit of the projection angle range, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE_MODE	Sets whether to consider projection angles within an angular range when extracting the profile to search. The chosen projection angle is determined based on the number of markers found and the average strength of those markers. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to use multiple projection angles.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to use multiple projection angles.	c		e	f	g	a		i		j	k	l	m	n	b

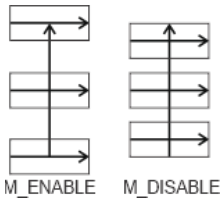
M_PROFILE_PROJECTION_ANGLE_TOLERANCE	Sets the tolerance used to determine the projection angles. The tolerance determines the spacing of the first phase of the angle refinement. The angle refinement is performed at the nominal projection angle and at every +/- M_PROFILE_PROJECTION_ANGLE_TOLERANCE degrees within the projection angle range. When M_PROFILE_PROJECTION_ANGLE_MODE is disabled, this setting has no effect. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 2.5°.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* <= 170.0	Specifies the tolerance, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_SAMPLE_SIZE	Sets the sampling distance between two consecutive points along the profile line. When the pixel size in X is not equal to the pixel size in Y in the source depth map, this ensures that the extracted points are at a uniform distance. Note that this will be the pixel size of the profile. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the distance. Use M_PROFILE_SAMPLE_SIZE_MODE to specify how to interpret the specified sampling distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_SAMPLE_SIZE_MODE	Sets how to interpret the specified sampling distance (M_PROFILE_SAMPLE_SIZE). The M_RELATIVE_TO_PIXEL_SIZE_... modes allow the sampling distance to be dependent on the depth map calibration. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_RELATIVE_TO_PIXEL_SIZE_MIN.	c		e	f	g	a		i		j	k	l	m	n	b
M_ABSOLUTE	Specifies to use the sampling distance directly, without applying a multiplying factor. For example, if you set M_PROFILE_SAMPLE_SIZE to 2.0, the sampling distance will be 2.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the sampling distance by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the sampling distance by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the sampling distance by the pixel size in X. For example, if you set M_PROFILE_SAMPLE_SIZE to 2.0 and the pixel size in X is 0.5 mm, the sampling distance will be 1.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b

	M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the sampling distance by the pixel size in Y. For example, if you set M_PROFILE_SAMPLE_SIZE to 2.0 and the pixel size in Y is 0.6 mm, the sampling distance will be 1.2 mm.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_THICKNESS		Sets the thickness of the profile line. Pixels along the line, within the specified thickness, are sampled and corresponding points are considered for the profile. Note that a thick profile line typically results in multiple points contributing to the same profile point. You can use M_PROFILE_ACCUMULATE_TYPE to determine how multiple point values are combined to set the value of the corresponding profile point. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the thickness. Use M_PROFILE_THICKNESS_MODE to specify how to interpret the specified thickness.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_THICKNESS_MODE		Sets how to interpret the specified thickness value (M_PROFILE_THICKNESS). The M_RELATIVE_TO_PIXEL_SIZE_... modes allow the thickness to be dependent on the depth map calibration. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_RELATIVE_TO_PIXEL_SIZE_MIN.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ABSOLUTE	Specifies to use the thickness value directly, without applying a multiplying factor. For example, if you set M_PROFILE_THICKNESS to 2.0, the profiles will be 2.0 mm thick.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the thickness value by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the thickness value by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the thickness value by the pixel size in X. For example, if you set M_PROFILE_THICKNESS to 2.0 and the pixel size in X is 0.5 mm, the profiles will be 1.0 mm thick.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the thickness value by the pixel size in Y. For example, if you set M_PROFILE_THICKNESS to 2.0 and the pixel size in Y is 0.6 mm, the profiles will be 1.2 mm thick.	c		e	f	g	a		i		j	k	l	m	n	b

	M_RELATIVE_TO_SPACING	Specifies to multiply the thickness value by the profile spacing. For example, if you set M_PROFILE_THICKNESS to 2.0 and the spacing between profiles is 2.0 mm, the profiles will be 4.0 mm thick. Note that this value is only available if a template 3D measurement context is specified and M_TEMPLATE_PROFILE_SPACING_MODE is not set to M_RELATIVE_TO_THICKNESS.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and ControlValue parameter settings can be specified for a template in a template 3D measurement context. PathOrTemplateIndex must be set to the index of a template, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To control a template in a template 3D measurement context																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_TEMPLATE_EXTENDED_SEARCH		Sets whether to extend the search outside of the template's defined area. When this control type is enabled, the profiles are positioned normally and all data within the profile's thickness, including data outside of the template, is considered. When this control type is disabled, the first profile will be offset towards the center of the template such that its thickness does not extend outside the template, meaning data outside of the template's defined area is never considered. INQ  															
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to extend the search beyond the template.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to extend the search beyond the template.	c		e	f	g	a		i		j	k	l	m	n	b

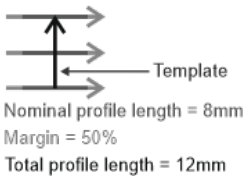
M_TEMPLATE_PROFILE_NUMBER_ELEMENTS	Sets the number of profile elements in the template 3D measurement context. Each profile element contains the settings for a single profile. If you reduce the number of profile elements, the settings of the ones that remain are kept unchanged. If you increase the number of profile elements, the new ones are initialized with the default settings. You should have at least as many profile elements as the number of profiles to extract. You configure profile elements by setting the M_TEMPLATE_PROFILE_..._SOURCE controls to M_PROFILE and setting the corresponding control in the To control a template or profile in a template 3D measurement context table. Note this value is not available when M_TEMPLATE_PROFILE_NUMBER_MODE is set to M_SPACING. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 512.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= M_TEMPLATE_PROFILE_NUMBER_VALUE	Specifies the number of profile elements.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_NUMBER_MODE	Sets the mode with which to determine the number of profiles to use to find the markers of a template. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_USER_DEFINED.	c		e	f	g	a		i		j	k	l	m	n	b
M_SPACING	Specifies that the number of profiles is determined based on the spacing between the profiles.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED	Specifies to use the value set with M_TEMPLATE_PROFILE_NUMBER_VALUE. The distance between profiles is established by distributing the specified number of profiles evenly perpendicular to the template.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_NUMBER_VALUE	Sets the number of profiles perpendicular to the template. The distance between profiles is established by evenly distributing the specified number of profiles. Note this value is only available when M_TEMPLATE_PROFILE_NUMBER_MODE is set to M_USER_DEFINED. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 10.	c		e	f	g	a		i		j	k	l	m	n	b
0 < *Value* <= M_TEMPLATE_PROFILE_NUMBER_ELEMENTS	Specifies the number of profiles.	c		e	f	g	a		i		j	k	l	m	n	b

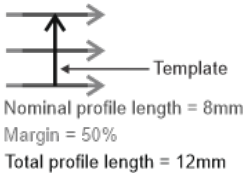
M_TEMPLATE_PROFILE_SPACING	Sets the spacing between the profiles along the template. Note that this control type is only used if M_TEMPLATE_PROFILE_NUMBER_MODE is set to M_SPACING. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the spacing. Use M_TEMPLATE_PROFILE_SPACING_MODE to specify how to interpret the specified spacing.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_SPACING_MODE	Sets how to interpret the specified spacing value (M_TEMPLATE_PROFILE_SPACING). The M_RELATIVE_TO_PIXEL_SIZE_... modes allow the spacing to be dependent on the depth map calibration. Note that this control type is only used if M_TEMPLATE_PROFILE_NUMBER_MODE is set to M_SPACING. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_RELATIVE_TO_THICKNESS.	c		e	f	g	a		i		j	k	l	m	n	b
M_ABSOLUTE	Specifies to use the spacing value directly, without applying a multiplying factor. For example, if you set M_TEMPLATE_PROFILE_SPACING to 2.0, the spacing between profiles will be 2.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the spacing value by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the spacing value by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the spacing value by the pixel size in X. For example, if you set M_TEMPLATE_PROFILE_SPACING to 2.0 and the pixel size in X is 0.5 mm, the spacing between profiles will be 1.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the spacing value by the pixel size in Y. For example, if you set M_TEMPLATE_PROFILE_SPACING to 2.0 and the pixel size in Y is 0.6 mm, the spacing between profiles will be 1.2 mm.	c		e	f	g	a		i		j	k	l	m	n	b

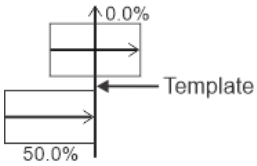
	M_RELATIVE_TO_THICKNESS	Specifies to multiply the spacing value by the profile thickness. For example, if you set M_TEMPLATE_PROFILE_SPACING to 2.0 and the profile thickness is 2.0 mm, the spacing between profiles will be 4.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and ControlValue parameter settings can be specified for a template or profile in a template 3D measurement context. PathOrTemplateIndex must be set to the index of a template. If ProfileIndex is set to M_DEFAULT, these control types allow you to specify global settings for the template's perpendicular profiles. If ProfileIndex is set to a specific index, they allow you to specify custom settings for the specified profile. Note that you must set the corresponding M_TEMPLATE_PROFILE_..._SOURCE control to M_PROFILE for the custom settings of the profile to be used.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To control a template or profile in a template 3D measurement context																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_TEMPLATE_PROFILE_LENGTH		Sets the nominal length of the profile to extract (perpendicular to the template). You should set the nominal length large enough to include the expected position variation of the marker. If you want to return a transition even when it is outside the expected range of positions, you can add a margin to the length, using M_TEMPLATE_PROFILE_LENGTH_MARGIN, and Aurora Imaging Library will continue the search within the margin.  Note that if you are setting the nominal length for a specific profile, you must set M_TEMPLATE_PROFILE_LENGTH_SOURCE to M_PROFILE to use the custom length value. INQ															
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the length. Use M_TEMPLATE_PROFILE_LENGTH_MODE to specify how to interpret the specified length.	c		e	f	g	a		i		j	k	l	m	n	b

M_TEMPLATE_PROFILE_LENGTH_MARGIN	Sets a margin on the profile length. You should include a margin when you still want to find a transition when it is located outside of the positions that you want to verify. The nominal length, specified with M_TEMPLATE_PROFILE_LENGTH, plus the specified margin define the length of the profile.  Note that if you are setting the margin for a specific profile, you must set M_TEMPLATE_PROFILE_LENGTH_MARGIN_SOURCE to M_PROFILE to use the custom margin value. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the margin, as a percentage of the nominal length.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_LENGTH_MODE	Sets how to interpret the specified length value (M_TEMPLATE_PROFILE_LENGTH). The M_RELATIVE_TO_PIXEL_SIZE_... modes allow the profile length to be dependent on the depth map calibration. Note that if you are setting the length mode for a specific profile, you must set M_TEMPLATE_PROFILE_LENGTH_SOURCE to M_PROFILE to use the custom length mode. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_RELATIVE_TO_PIXEL_SIZE_MIN.	c		e	f	g	a		i		j	k	l	m	n	b
M_ABSOLUTE	Specifies to use the length value directly, without applying a multiplying factor. For example, if you set M_TEMPLATE_PROFILE_LENGTH to 2.0, the profile length will be 2.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the length value by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the length value by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the length value by the pixel size in X. For example, if you set M_TEMPLATE_PROFILE_LENGTH to 2.0 and the pixel size in X is 0.5 mm, the profile length will be 1.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the length value by the pixel size in Y. For example, if you set M_TEMPLATE_PROFILE_LENGTH to 2.0 and the pixel size in Y is 0.6 mm, the profile length will be 1.2 mm.	c		e	f	g	a		i		j	k	l	m	n	b

M_TEMPLATE_PROFILE_POSITION_RELATIVE		Sets the position of the profile, relative to the template. By default, the profiles are centered on the template. You must specify the position as a percentage of the nominal length of the profile (not including the margin). For example, if a profile has a nominal length of 10 mm and you specify a value of 50.0%, the profile is moved 5 mm backwards, such that the end of its nominal length touches the template; if a margin was specified, it will continue across the template. If you specify a value of -50.0%, the profile is moved 5 mm forwards, such that the start of its nominal length touches the template.  Note that if you are setting the relative position for a specific profile, you must set M_TEMPLATE_PROFILE_POSITION_SOURCE to M_PROFILE to use the custom relative position value. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	-50.0 <= *Value* <= 50.0	Specifies the relative position, as a percentage of the nominal length.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and ControlValue parameter settings can be specified for a profile perpendicular to a template in a template 3D measurement context. PathOrTemplateIndex must be set to the index of a template, and ProfileIndex must be set to the index of one or all profiles. These control types allow you to specify whether to use the global or custom settings for the specified profile, when extracting the profile perpendicular to the template.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To control a profile in a template 3D measurement context																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_TEMPLATE_PROFILE_LENGTH_MARGIN_SOURCE		Sets whether to use the global margin value or a custom margin for the profile. The margin is added to the nominal length to define the length of the profile. INQ															
	M_DEFAULT	Same as M_TEMPLATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PROFILE	Specifies to use the custom margin value for the specified profile. You must set the custom margin using M_TEMPLATE_PROFILE_LENGTH_MARGIN with ProfileIndex set to the index of the profile.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TEMPLATE	Specifies to use the template's global margin value for the specified profile. You must set the template's global margin using M_TEMPLATE_PROFILE_LENGTH_MARGIN with ProfileIndex set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_LENGTH_SOURCE		Sets whether to use the global nominal length value and length mode or a custom nominal length and length mode for the profile. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_TEMPLATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PROFILE	Specifies to use the custom profile length for the specified profile. You must set the custom length using M_TEMPLATE_PROFILE_LENGTH and M_TEMPLATE_PROFILE_LENGTH_MODE with ProfileIndex set to the index of the profile.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TEMPLATE	Specifies to use the template's global profile length for the specified profile. You must set the template's global length using M_TEMPLATE_PROFILE_LENGTH and M_TEMPLATE_PROFILE_LENGTH_MODE with ProfileIndex set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_POSITION_SOURCE		Sets whether to use the global relative position value or a custom relative position for the profile. The relative position is the position of the profile relative to the template. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_TEMPLATE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PROFILE	Specifies to use the custom relative position value for the specified profile. You must set the custom relative position using M_TEMPLATE_PROFILE_POSITION_RELATIVE with ProfileIndex set to the index of the profile.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TEMPLATE	Specifies to use the template's global relative position value. You must set the template's global relative position using M_TEMPLATE_PROFILE_POSITION_RELATIVE with ProfileIndex set to M_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and ControlValue parameter settings can be specified for a fit 3D measurement context. PathOrTemplateIndex must be set to M_DEFAULT or M_GENERAL, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To control a fit 3D measurement context																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																

M_EXPECTED_OUTLIER_PERCENTAGE	Sets the expected percentage of outlier markers among the markers that are extracted. INQ														
M_DEFAULT	Specifies the default value; the default value is 20.0%.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 < *Value* < 100.0	Specifies the expected outliers, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_COVERAGE_MIN	Sets the minimum percentage of profiles that must have a marker covered by a fitted point. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0%.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the minimum coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE	Sets the distance value that Aurora Imaging Library uses to establish outlier markers, which are excluded from the robust best fit. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies that the fit distance is infinite. When M_INFINITE is specified, no markers are considered outliers, so no markers are excluded from the robust best fit.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE_MODE	Sets how to interpret the specified fit distance value (M_FIT_DISTANCE). The M_RELATIVE_TO_PIXEL_SIZE_... modes allow the fit distance to be dependent on the depth map calibration. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_AUTO.	c		e	f	g	a		i		j	k	l	m	n	b
M_ABSOLUTE	Specifies to use the fit distance value directly, without applying a multiplying factor. For example, if you set M_FIT_DISTANCE to 2.0, the fit distance will be 2.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically calculate the fit distance; the distance is estimated from all the extracted markers. Note that when the find and fit are performed together (by calling M3dmeasFit() with a template 3D measurement context and a depth map), and there are multiple valid transitions within a single profile, the automatic fit distance can be overestimated. In this case, you should use a different mode.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the fit distance value by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b

	M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the fit distance value by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the fit distance value by the pixel size in X. For example, if you set M_FIT_DISTANCE to 2.0 and the pixel size in X is 0.5 mm, the fit distance will be 1.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the fit distance value by the pixel size in Y. For example, if you set M_FIT_DISTANCE to 2.0 and the pixel size in Y is 0.6 mm, the fit distance will be 1.2 mm.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_SAMPLING	Specifies to multiply the fit distance value by the sampling distance along each profile. For example, if you set M_FIT_DISTANCE to 2.0 and the sampling distance along each profile is 2.0 mm, the fit distance will be 4.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_OUTLIERS		Sets whether to remove outlier markers from the template 3D measurement result buffer after the fit. Note that when the find and fit are performed simultaneously, using M3dmeasFit() with a template 3D measurement context and a depth map, the outliers are always removed, even if this control type is set to M_DISABLE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to remove outliers.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to remove outliers.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and ControlValue parameter setting can be specified for a profile, path, or template 3D measurement result buffer. PathOrTemplateIndex must be set to M_DEFAULT or M_GENERAL, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To control a profile, path, or template 3D measurement result buffer																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_STOP_FIND		Stops the current execution of M3dmeasFindMarker() or M3dmeasFit(). This can only be done from another thread of higher priority. Results already available in the result buffer become invalid and will be discarded.															
	M_DEFAULT	Specifies the default behavior.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void M3dmeasControlInt64 (AIL_ID ContextOrResult3dmeasId, AIL_INT64 PathOrTemplateIndex, AIL_INT64 ProfileIndex,
AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

ContextOrResult3dmeasId

See ContextOrResult3dmeasId of the main function for a description.

PathOrTemplateIndex

See PathOrTemplateIndex of the main function for a description.

ProfileIndex

See ProfileIndex of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void M3dmeasControlDouble (AIL_ID ContextOrResult3dmeasId, AIL_INT64 PathOrTemplateIndex, AIL_INT64 ProfileIndex,
AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

ContextOrResult3dmeasId

See ContextOrResult3dmeasId of the main function for a description.

PathOrTemplateIndex

See PathOrTemplateIndex of the main function for a description.

ProfileIndex

See ProfileIndex of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasControlDraw

Synopsis

Control a setting of a draw 3D measurement context.

Syntax

```
void M3dmeasControlDraw(
    AIL_ID DrawContext3dmeasId, //out
    AIL_INT64 Operation,        //in
    AIL_INT64 ControlType,      //in
    AIL_DOUBLE ControlValue     //in
)
```

Description

This function controls a specified setting of a draw 3D measurement context. These settings establish which 3D measurement results to draw into the 3D graphics list (and how to draw them), when calling M3dmeasDraw3d().

You can inquire about most of these settings using M3dmeasInquireDraw().

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **M3dmeasControlDraw()** internally calls the *AIL_DOUBLE* version of this function (M3dmeasControlDrawDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (M3dmeasControlDrawInt64()).

Parameters

DrawContext3dmeasId

Specifies the identifier of the draw 3D measurement context to control. The 3D measurement context must have been previously allocated on the system using M3dmeasAlloc() with M_DRAW_3D_PROFILE_CONTEXT, M_DRAW_3D_PATH_CONTEXT, or M_DRAW_3D_TEMPLATE_CONTEXT.

Operation

Specifies the draw operation.

See the Parameter associations section for possible values that can be specified.

ControlType

Specifies the setting to change.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the setting's new value.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The tables below list possible values for the **Operation**, **ControlType**, and **ControlValue** parameters.

- For controlling a draw profile, path, or template 3D measurement context setting
- For controlling a draw path 3D measurement context setting
- For controlling a draw template 3D measurement context setting

The following Operation, ControlType, and ControlValue parameter settings are available to control draw profile, path, or template 3D measurement context settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling a draw profile, path, or template 3D measurement context setting

Operation / ControlType / ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ALL	Applies the setting to all operations that support the control type.															
M_ACTIVE	Sets whether to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets the appearance of the graphic(s) to solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

			Red Specifies the red band, as a value between 0 and 255.																
			Green Specifies the green band, as a value between 0 and 255.																
			Blue Specifies the blue band, as a value between 0 and 255.																
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies to use the default setting of the operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MARKERS	Sets whether and how to draw the cross-like symbol at each specified marker's position, and for pair markers, a rectangle connecting the two transitions with a cross-like symbol at each transition position, when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINES. Note that this is equivalent to M_DRAW_MARKERS_POSITION for single markers and M_DRAW_MARKERS_POSITION + M_DRAW_MARKERS_WIDTH for pair markers.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. Note that a cross-like symbol drawn with a solid appearance is the same as a cross-like symbol drawn with a wireframe appearance. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_PSEUDO_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b

M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MARKERS_POSITION	Sets whether and how to draw the cross-like symbol at each specified marker's position when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINES. Note that this is equivalent to M_DRAW_MARKERS for single markers.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. Note that a cross-like symbol drawn with a solid appearance is the same as a cross-like symbol drawn with a wireframe appearance. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b

M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_PSEUDO_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MARKERS_WIDTH	Sets whether and how to draw the width of each specified pair marker when calling M3dmeasDraw3d(). The width is drawn as a rectangle connecting the two transitions with a cross-like symbol at each transition position. The resulting graphic will be of type M_GRAPHIC_TYPE_LINES. This operation is only supported for pair markers.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. Note that a cross-like symbol drawn with a solid appearance is the same as a cross-like symbol drawn with a wireframe appearance. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_PSEUDO_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b

		M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
			Parameters															
			Red Specifies the red band, as a value between 0 and 255.															
			Green Specifies the green band, as a value between 0 and 255.															
			Blue Specifies the blue band, as a value between 0 and 255.															
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PROFILE_EXTRACTED	Sets whether and how to draw the extracted points that form each specified profile when calling M3dmeasDraw3d(). Note that this includes points introduced after filling gaps. The resulting graphic will be of type M_GRAPHIC_TYPE_LINES.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_YELLOW.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PROFILE_REGION	Sets whether and how to draw the region where each specified profile was extracted when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_POLYGON.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_MAGENTA.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 20.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b

M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PROFILE_RESPONSE	Sets whether and how to draw the response of each specified profile (after applying the filter) when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINES.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_CYAN.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b

			Parameters																
			Red Specifies the red band, as a value between 0 and 255.																
			Green Specifies the green band, as a value between 0 and 255.																
			Blue Specifies the blue band, as a value between 0 and 255.																
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_TRANSITIONS	Sets whether and how to draw the step-like symbol at each specified marker's transition position when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINES.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_PSEUDO_DARK_COLOR.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b

M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 1 pixel. Specifies the default value; the default value is 1 pixel.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_DRAW_SETTINGS	Sets a global draw 3D measurement context setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR_OFFSET	Sets the offset to apply to either the 3D measurement result element's index when drawing with M_PSEUDO_COLOR or M_PSEUDO_DARK_COLOR. You can use this control type to prevent the same colors from being assigned to different result elements when drawing several results at once. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the offset to apply to the index.	c		e	f	g	a		i		j	k	l	m	n	b

The following Operation, ControlType, and ControlValue parameter settings are available to control draw path 3D measurement context settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling a draw path 3D measurement context setting																
Operation ControlType ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_PATH	Sets whether and how to draw the path defined in the path 3D measurement context when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINE.															
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_MAGENTA.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PATH_PROJECTED	Sets whether and how to draw the projection of the path defined in the path 3D measurement context when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_POLYGON.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_MAGENTA.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b

		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

The following Operation, ControlType, and ControlValue parameter settings are available to control draw template 3D measurement context settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For controlling a draw template 3D measurement context setting																		
Operation ControlType ControlValue	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_FIND_GEOMETRY	Sets whether and how to draw the line segments connecting the extracted markers when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINES.															
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b

M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_CYAN.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_FIND_GEOMETRY_PROJECTED	Sets whether and how to draw the planes connecting the projections of the extracted markers when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_POINT_CLOUD.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_DARK_CYAN.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

			Red Specifies the red band, as a value between 0 and 255.																
			Green Specifies the green band, as a value between 0 and 255.																
			Blue Specifies the blue band, as a value between 0 and 255.																
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 50.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIT_GEOMETRY			Sets whether and how to draw the fitted 3D geometry when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE		Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_YELLOW.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIT_GEOMETRY_PROJECTED	Sets whether and how to draw the projection of the fitted 3D geometry when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_POLYGON.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b

	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR		Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_COLOR_DARK_YELLOW.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
		Parameters															
		Red Specifies the red band, as a value between 0 and 255.															
		Green Specifies the green band, as a value between 0 and 255.															
		Blue Specifies the blue band, as a value between 0 and 255.															
	M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 50.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIT_MARKERS	Sets whether and how to draw the cross-like symbol at the position of each marker that was used for the fit operation when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINES.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. Note that a cross-like symbol drawn with a solid appearance is the same as a cross-like symbol drawn with a wireframe appearance. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b

M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_GREEN.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_FIT_MARKERS_OUTLIERS	Sets whether and how to draw the cross-like symbol at the position of each marker that was considered an outlier for the fit operation when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINES.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. Note that a cross-like symbol drawn with a solid appearance is the same as a cross-like symbol drawn with a wireframe appearance. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_RED.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

			Red Specifies the red band, as a value between 0 and 255.																
			Green Specifies the green band, as a value between 0 and 255.																
			Blue Specifies the blue band, as a value between 0 and 255.																
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS		Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_TEMPLATE			Sets whether and how to draw the template defined in the template 3D measurement context when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_LINE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE		Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR	Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_COLOR_MAGENTA.	c		e	f	g	a		i		j	k	l	m	n	b
M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Red Specifies the red band, as a value between 0 and 255.															
	Green Specifies the green band, as a value between 0 and 255.															
	Blue Specifies the blue band, as a value between 0 and 255.															
M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY	Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 100.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_TEMPLATE_PROJECTED	Sets whether and how to draw the projection of the template defined in the template 3D measurement context when calling M3dmeasDraw3d(). The resulting graphic will be of type M_GRAPHIC_TYPE_POLYGON.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE	Sets whether to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE. Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE	Sets whether to draw the graphic(s) as a solid surface, wireframe, or points. The color of the points, wireframe, or outline is determined by M_COLOR, while the color of the solid surface is determined by the default fill color of the 3D graphics list (M3dgraControl() with M_FILL_COLOR). INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b

		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_POINTS	Specifies a points appearance. The graphic(s) appear as a set of points representing its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID	Specifies a solid appearance.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_POINTS	Specifies a solid appearance with points.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe.	c		e	f	g	a		i		j	k	l	m	n	b
		M_WIREFRAME	Specifies a wireframe appearance. The graphic(s) appear as a set of lines connecting its vertices.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR		Sets the color with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_COLOR_MAGENTA.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RGB888(AIL_INT Red, AIL_INT Green, AIL_INT Blue)	Specifies the RGB value to use as the color.	c		e	f	g	a		i		j	k	l	m	n	b
			Parameters															
			Red Specifies the red band, as a value between 0 and 255.															
			Green Specifies the green band, as a value between 0 and 255.															
			Blue Specifies the blue band, as a value between 0 and 255.															
		M_COLOR_BLACK	Specifies the color black.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green.	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY		Sets the opacity with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value; the default value is 50.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_THICKNESS	Sets the thickness with which to draw the graphic(s). INQ	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the default value. Same as M_GRAPHIC_LIST_DEFAULT.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void M3dmeasControlDrawDouble (AIL_ID DrawContext3dmeasId, AIL_INT64 Operation, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

DrawContext3dmeasId

See DrawContext3dmeasId of the main function for a description.

Operation

See Operation of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void M3dmeasControlDrawInt64 (AIL_ID DrawContext3dmeasId, AIL_INT64 Operation, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

DrawContext3dmeasId

See DrawContext3dmeasId of the main function for a description.

Operation

See Operation of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.

Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasCopy

Synopsis

Copy a path or template to a 3D geometry.

Syntax

```
void M3dmeasCopy(
    AIL_ID SrcContext3dmeasId, //in
    AIL_INT64 SrcIndex,        //in
    AIL_ID DstAilObjectId,     //out
    AIL_INT64 DstIndex,        //in
    AIL_INT64 CopyType,        //in
    AIL_INT64 ControlFlag      //in
)
```

Description

This function copies a path in a path 3D measurement context or a template in a template 3D measurement context to a 3D geometry.

Parameters

SrcContext3dmeasId

Specifies the identifier of the source 3D measurement context from which to copy.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the source 3D measurement context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
Path 3D measurement context identifier	Specifies the identifier of the path 3D measurement context from which to copy the path.	c		e	f	g	a		i		j	k	l	m	n	b
Template 3D measurement context identifier	Specifies the identifier of the template 3D measurement context from which to copy the template.	c		e	f	g	a		i		j	k	l	m	n	b

SrcIndex

Specifies the index of the path or template in the source 3D measurement context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a path or template																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_PATH_INDEX(AIL_INT PathIndex)	Specifies the path in a path 3D measurement context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PathIndex Set this parameter to the following:															

		0 Specifies the index of the path.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies the template in a template 3D measurement context, if one is specified.		c		e	f	g	a		i		j	k	l	m	n	b
	Parameters																
	TemplateIndex Set this parameter to the following:																
		0 Specifies the index of the template.	c		e	f	g	a		i		j	k	l	m	n	b

DstAilObjectId

Specifies the identifier of the 3D geometry object in which to copy the path or template.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the destination 3D geometry object																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
3D geometry object identifier	Specifies the identifier of the 3D geometry object in which to copy the path or template. The 3D geometry object must have been previously allocated using M3dgeoAlloc() with M_GEOMETRY.	c		e	f	g	a		i		j	k	l	m	n	b

DstIndex

Reserved for future expansion and must be set to **M_DEFAULT**.

CopyType

Specifies the type of copy operation to perform.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

Specifies the type of copy operation																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_GEOMETRY_PATH	Specifies to copy the path from the path 3D measurement context into the specified 3D geometry object, establishing a 3D line geometry.	c		e	f	g	a		i		j	k	l	m	n	b
M_GEOMETRY_TEMPLATE	Specifies to copy the template from the template 3D measurement context into the specified 3D geometry object, establishing a 3D line geometry.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasCopyResult

Synopsis

Copy a group of results from a 3D measurement result buffer into a 3D geometry object, container, or image buffer.

Syntax

```
void M3dmeasCopyResult(
    AIL_ID SrcResult3dmeasId,         //in
    AIL_INT64 SrcPathOrTemplateIndex, //in
    AIL_INT64 SrcProfileIndex,        //in
    AIL_ID DstAilObjectId,            //out
    AIL_INT64 DstIndex,               //in
    AIL_INT64 CopyType,               //in
    AIL_INT64 ControlFlag             //in
)
```

Description

This function copies a group of results from a 3D measurement result buffer into a 3D geometry object, container, or image buffer.

Parameters

SrcResult3dmeasId

Specifies the identifier of a 3D measurement result buffer from which to copy results. The 3D measurement result buffer must have been allocated using M3dmeasAllocResult().

See the Parameter associations section for possible values that can be specified.

SrcPathOrTemplateIndex

Specifies the index of the source template or path to copy.

See the Parameter associations section for possible values that can be specified.

SrcProfileIndex

Specifies the index of the source profile to copy.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

DstAilObjectId

Specifies the identifier of the image buffer, 3D geometry object, or container in which to store the copied results. The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must not be a child container.

See the Parameter associations section for possible values that can be returned.

DstIndex

Reserved for future expansion and must be set to **M_DEFAULT**.

CopyType

Specifies the type of copy operation to perform.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Parameter associations

The table below lists possible values for the **SrcResult3dmeasId**, **SrcPathOrTemplateIndex**, **SrcProfileIndex**, and **CopyType** parameters and possible values returned to the **DstAilObjectId** parameter.

- For specifying the copy type

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the copy type																		
CopyType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
SrcResult3dmeasId																
SrcPathOrTemplateIndex																
SrcProfileIndex																
DstAilObjectId - Possible values returned																
M_GEOMETRY_FIT	Specifies to copy the fitted geometry from the specified template 3D measurement result buffer into the specified 3D geometry object. Note that this is only available after an M3dmeasFit() operation.															
SrcResult3dmeasId	Specifies the identifier of the template 3D measurement result buffer from which to copy the fitted geometry.	c		e	f	g	a		i		j	k	l	m	n	b
SrcPathOrTemplateIndex	Specifies the index of the template in the template 3D measurement result buffer for which to copy the fitted geometry.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies the index of the template.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	TemplateIndex Set this parameter to the following:															
	0 — Specifies the index.	c		e	f	g	a		i		j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of the destination 3D geometry object in which to copy the fitted geometry.	c		e	f	g	a		i		j	k	l	m	n	b
M_GEOMETRY_PATH	Specifies to copy the path used to calculate the results in the specified path 3D measurement result buffer, into the specified 3D geometry object.	c		e	f	g	a		i		j	k	l	m	n	b
SrcResult3dmeasId	Specifies the identifier of the path 3D measurement result buffer from which to copy the path.	c		e	f	g	a		i		j	k	l	m	n	b
SrcPathOrTemplateIndex	Specifies the index of the path to copy from the path 3D measurement result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

M_PATH_INDEX(AIL_INT PathIndex)	Specifies the index of the path.	c	e	f	g	a	i	j	k	l	m	n	b
	Parameters												
	PathIndex Set this parameter to the following:												
	0 Specifies the index.	c	e	f	g	a	i	j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of the destination 3D geometry object in which to copy the path.	c	e	f	g	a	i	j	k	l	m	n	b
M_GEOMETRY_TEMPLATE	Specifies to copy the template used to calculate the results in the specified template 3D measurement result buffer, into the specified 3D geometry object.	c	e	f	g	a	i	j	k	l	m	n	b
SrcResult3dmeasId	Specifies the identifier of the template 3D measurement result buffer from which to copy the template.	c	e	f	g	a	i	j	k	l	m	n	b
SrcPathOrTemplateIndex	Specifies the index of the template to copy from the template 3D measurement result buffer.	c	e	f	g	a	i	j	k	l	m	n	b
M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies the index of the template.	c	e	f	g	a	i	j	k	l	m	n	b
	Parameters												
	TemplateIndex Set this parameter to the following:												
	0 Specifies the index.	c	e	f	g	a	i	j	k	l	m	n	b
DstAilObjectId	Specifies the identifier of the destination 3D geometry object in which to copy the template.	c	e	f	g	a	i	j	k	l	m	n	b
M_PROFILE_EXTRACTED	Specifies to copy the extracted profile from the specified 3D measurement result buffer into the first row of the specified image buffer or container.	c	e	f	g	a	i	j	k	l	m	n	b
SrcResult3dmeasId	Specifies the identifier of the path or template 3D measurement result buffer from which to copy the profile.	c	e	f	g	a	i	j	k	l	m	n	b
SrcPathOrTemplateIndex	Specifies the index of the path or template in the path or template 3D measurement result buffer that has the profile to copy.	c	e	f	g	a	i	j	k	l	m	n	b
M_PATH_INDEX(AIL_INT PathIndex)	Specifies the index of the path in the path 3D measurement result buffer.	c	e	f	g	a	i	j	k	l	m	n	b

			Parameters														
			PathIndex Set this parameter to the following:														
			0 Specifies the index.	c	e	f	g	a	i	j	k	l	m	n	b		
		M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies the index of the template in the template 3D measurement result buffer.	c	e	f	g	a	i	j	k	l	m	n	b		
			Parameters														
			TemplateIndex Set this parameter to the following:														
			0 Specifies the index.	c	e	f	g	a	i	j	k	l	m	n	b		
	SrcProfileIndex		Specifies the index of the profile in the 3D measurement result buffer to copy.	c	e	f	g	a	i	j	k	l	m	n	b		
		Value >= 0	Specifies the index of a profile. Note that for a path 3D measurement result buffer, only index 0 is supported because only 1 profile is extracted along the path.	c	e	f	g	a	i	j	k	l	m	n	b		
	DstAilObjectId		Specifies the identifier of the destination image buffer or container in which to copy the profile.	c	e	f	g	a	i	j	k	l	m	n	b		
		Container identifier	Specifies the identifier of a container. The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must not be a child container. Note that after the profile is copied, the specified container will be a 3D-processable depth map container with a single row.	c	e	f	g	a	i	j	k	l	m	n	b		
		Image buffer identifier	Specifies the identifier of an image buffer. The image buffer must be an 8-bit, 16-bit, or 32-bit unsigned, 1-band buffer. Note that the profile is copied into the first row of the specified image buffer.	c	e	f	g	a	i	j	k	l	m	n	b		

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasDefine

Synopsis

Add a path or template to, or delete a path or template from, a path or template 3D measurement context.

Syntax

```
AIL_INT64 M3dmeasDefine(
    AIL_ID PathOrTemplateContext3dmeasId, //out
    AIL_ID Geometry3dgeoId,               //in
    AIL_ID Matrix3dgeoId,                 //in
    AIL_INT PathOrTemplateIndex,          //in
    AIL_INT64 Operation,                  //in
    AIL_INT MarkerType,                   //in
    AIL_INT TransitionType,               //in
    AIL_INT NumberMode,                   //in
    AIL_INT LengthSource,                 //in
    AIL_INT LengthMode,                   //in
    AIL_INT NumberValue,                  //in
    const AIL_DOUBLE *ParamArrayPtr,      //in
    AIL_INT64 ControlFlag                 //in
)
```

Description

This function allows you to add or delete the path or template that is used to find markers. Note that a path can only be added to, or deleted from, a path 3D measurement context, and a template can only be added to, or deleted from, a template 3D measurement context.

Setting a path is useful when you want to find an object but you don't know its exact location. You can set a path that crosses through possible locations to detect height changes along the line. Setting a template is useful when you want to check whether an object is at an expected location. You can set a template where you expect a height change to occur, and verify if the object is present and has the shape you expect. You can use M3dmeasFit() to fit a geometry to the found markers to compare it with the template that you defined.

You can only define one path per path 3D measurement context and one template per template 3D measurement context. When adding a path or template to the context, it is assigned an index of 0. Note that you cannot add a path or template to a profile 3D measurement context. A profile 3D measurement context has a default template that corresponds to the profiles supplied to M3dmeasFindMarker() and cannot be defined nor deleted.

After defining the path or template, use M3dmeasFindMarker() to find markers along the path or find the markers that can form the template.

Parameters

PathOrTemplateContext3dmeasId

Specifies the path or template 3D measurement context to which to add, or from which to delete, the path or template. The context must have been allocated using M3dmeasAlloc() with M_FIND_MARKER_PATH_CONTEXT or M_FIND_MARKER_TEMPLATE_CONTEXT.

See the Parameter associations section for possible values that can be returned.

Geometry3dgeoId

Specifies the identifier of the 3D geometry object to use to define the path or template. The 3D geometry object must have been allocated using M3dgeoAlloc() with M_GEOMETRY and successfully defined as a line.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Matrix3dgeoId

Specifies the identifier of the transformation matrix object to apply to the path or template.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

PathOrTemplateIndex

Specifies the index of the template or path in the 3D measurement context.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

Operation

Specifies the operation to perform.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of operation																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ADD	Specifies to add a path or template to the specified 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DELETE	Specifies to delete the specified path or template from the specified 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b

MarkerType

Reserved for future expansion and must be set to **M_DEFAULT**.

TransitionType

Specifies the type of transition to find.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

NumberMode

Specifies the mode with which to determine the number of profiles that will be used to find the markers of a template.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

LengthSource

Specifies the source to use to define the length of the profiles.

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

LengthMode

Specifies the mode with which to interpret the specified length (ParamArrayPtr).

Set this parameter to **M_DEFAULT** if not used.

See the Parameter associations section for possible values that can be specified.

NumberValue

Specifies the number of profiles.

Set this parameter to **M_DEFAULT** if not used.

> When using a standard vector (std::vector) overload function in C++, you can pass **M_DEFAULT** to this parameter; the size will be determined automatically based on the number of items in the vector passed to the ParamArrayPtr parameter.

See the Parameter associations section for possible values that can be specified.

ParamArrayPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>]
- AIL_DOUBLE
- M_NULL

Specifies the address of the array that contains the lengths of the profiles. If this information is not required, set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Parameter associations

The tables below list possible values for the **Geometry3dgeoId**, **Matrix3dgeoId**, **PathOrTemplateIndex**, **TransitionType**, **NumberMode**, **LengthSource**, **LengthMode**, **NumberValue**, and **ParamArrayPtr** parameters and possible values returned to the **PathOrTemplateContext3dmeasId** parameter.

- For adding a path or template to the 3D measurement context
- For deleting a path or template from the 3D measurement context

To add a path or template to the 3D measurement context, the PathOrTemplateContext3dmeasId, Geometry3dgeoId, Matrix3dgeoId, PathOrTemplateIndex, TransitionType, NumberMode, LengthSource, LengthMode, NumberValue, and ParamArrayPtr parameters can be set to the following values. The Operation parameter must be set to M_ADD.

Note that any unused parameters should be set to **M_DEFAULT**. Note, if unused, set ParamArrayPtr to **M_NULL**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For adding a path or template to the 3D measurement context			(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
PathOrTemplateContext3dmeasId - Possible values returned		Description															
	Geometry3dgeoId																
	Matrix3dgeoId																
	PathOrTemplateIndex																
	TransitionType																
	NumberMode																
	LengthSource																
	LengthMode																
	NumberValue																
	ParamArrayPtr																
Path 3D measurement context identifier in which to define a path		Specifies to add a path to a path 3D measurement context.															

	Geometry3dgeoId	Specifies the identifier of the 3D geometry object to use to define the path. The 3D geometry object must have been allocated using M3dgeoAlloc() with M_GEOMETRY and successfully defined as a line.	c		e	f	g	a		i		j	k	l	m	n	b
	Matrix3dgeoId	Specifies the identifier of the transformation matrix object to apply to the path. This allows you to fixture the path to a required position or orientation. For example, if you have identified the location of a subpart, you can apply the transformation matrix required for the path to be at a particular offset from that location. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX.	c		e	f	g	a		i		j	k	l	m	n	b
	PathOrTemplateIndex	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
	TransitionType	Specifies the type of transition to find.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_EDGE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGE	Specifies to search for edge transitions.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGE_OR_INVALID	Specifies to search for edge transitions and invalid transitions.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INVALID	Specifies to search for invalid transitions.	c		e	f	g	a		i		j	k	l	m	n	b
Template 3D measurement context identifier in which to define a template		Specifies to add a template to a template 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
	Geometry3dgeoId	Specifies the identifier of the 3D geometry object to use to define the template. The 3D geometry object must have been allocated using M3dgeoAlloc() with M_GEOMETRY and successfully defined as a line.	c		e	f	g	a		i		j	k	l	m	n	b

	Matrix3dgeoId	Specifies the identifier of the transformation matrix object to apply to the template. This allows you to fixture the template to a required position or orientation. For example, if you have identified the location of a subpart, you can apply the transformation matrix required for the template to be at a particular offset from that location. The transformation matrix object must have been previously allocated using M3dgeoAlloc() with M_TRANSFORMATION_MATRIX.	c		e	f	g	a		i		j	k	l	m	n	b
	PathOrTemplateIndex	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
	TransitionType	Specifies the type of transition to find.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_EDGE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGE	Specifies to search for edge transitions.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGE_OR_INVALID	Specifies to search for edge transitions and invalid transitions.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INVALID	Specifies to search for invalid transitions.	c		e	f	g	a		i		j	k	l	m	n	b
	NumberMode	Specifies the mode with which to determine the number of profiles that will be used to find the markers of a template.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_USER_DEFINED.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SPACING	Specifies that the number of profiles is based on the spacing between the profiles. The profiles are generated such that they are evenly spaced and span the entire template. Note that you must set the required spacing using M3dmeasControl() with M_TEMPLATE_PROFILE_SPACING and M_TEMPLATE_PROFILE_SPACING_MODE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED	Specifies that the number of profiles is explicitly set using the NumberValue parameter.	c		e	f	g	a		i		j	k	l	m	n	b
	LengthSource	Specifies the source to use to define the length of the profiles.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_TEMPLATE.	c		e	f	g	a		i		j	k	l	m	n	b

		M_PROFILE	Specifies that the length of each profile is set individually. You must set the length of each profile using ParamArrayPtr. Note that this value is only available when NumberMode is set to M_USER_DEFINED.	c		e	f	g	a		i		j	k	l	m	n	b
		M_TEMPLATE	Specifies that the length of each profile is determined by a single value set for the template. You must set the length of the profiles using ParamArrayPtr.	c		e	f	g	a		i		j	k	l	m	n	b
	LengthMode		Specifies the mode with which to interpret the specified length (ParamArrayPtr). The M_RELATIVE_TO_PIXEL_SIZE_... modes allow the profile length to be dependent on the depth map calibration.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Same as M_RELATIVE_TO_PIXEL_SIZE_MIN.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ABSOLUTE	Specifies to use the length value directly, without applying a multiplying factor. For example, if you set the length value to 2.0, the profile length will be 2.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the length value by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the length value by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the length value by the pixel size in X. For example, if you set the length value to 2.0 and the pixel size in X is 0.5 mm, the profile length will be 1.0 mm.	c		e	f	g	a		i		j	k	l	m	n	b
		M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the length value by the pixel size in Y. For example, if you set the length value to 2.0 and the pixel size in Y is 0.6 mm, the profile length will be 1.2 mm.	c		e	f	g	a		i		j	k	l	m	n	b
	NumberValue		Specifies the number of profiles.	c		e	f	g	a		i		j	k	l	m	n	b

		M_DEFAULT	Specifies the default value. If NumberMode is set to M_SPACING, specifies that the number value is not required. If NumberMode is set to M_USER_DEFINED, the default value is 10.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0	Specifies the number of profiles. Note that this value is only available when NumberMode is set to M_USER_DEFINED.	c		e	f	g	a		i		j	k	l	m	n	b
	ParamArrayPtr		Specifies the address of the array that contains the lengths of the profiles. When the LengthSource parameter is set to M_PROFILE, the number of elements in the array should be equal to the NumberValue parameter. When it is set to M_TEMPLATE, the array size should be 1. Alternatively, you can set this parameter to **M_NULL** to use the default length of the template 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b

To delete a path or template from the 3D measurement context, the PathOrTemplateContext3dmeasId, Geometry3dgeoId, Matrix3dgeoId, and PathOrTemplateIndex parameters can be set to the following values. The Operation parameter must be set to M_DELETE.

Note that any unused parameters should be set to **M_DEFAULT**. Note, set ParamArrayPtr to **M_NULL**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For deleting a path or template from the 3D measurement context																		
PathOrTemplateContext3dmeasId - Possible values returned		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	Geometry3dgeoId																	
	Matrix3dgeoId																	
	PathOrTemplateIndex																	
	TransitionType																	
	NumberMode																	
	LengthSource																	
	LengthMode																	
	NumberValue																	
	ParamArrayPtr																	
Path 3D measurement context identifier in which to remove a path		Specifies to delete a path from a path 3D measurement context.																
	Geometry3dgeoId	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b

Matrix3dgeoId	This parameter must be set to **M_DEFAULT**.	c	e	f	g	a	i	j	k	l	m	n	b
PathOrTemplateIndex	Specifies the index of the path in the path 3D measurement context.	c	e	f	g	a	i	j	k	l	m	n	b
M_PATH_INDEX(AIL_INT PathIndex)	Specifies the path to delete from the path 3D measurement context.	c	e	f	g	a	i	j	k	l	m	n	b
	Parameters												
	PathIndex Set this parameter to the following:												
	0 — Specifies the index of the path.	c	e	f	g	a	i	j	k	l	m	n	b
Template 3D measurement context identifier in which to remove a template	Specifies to delete a template from a template 3D measurement context.	c	e	f	g	a	i	j	k	l	m	n	b
Geometry3dgeoId	This parameter must be set to **M_DEFAULT**.	c	e	f	g	a	i	j	k	l	m	n	b
Matrix3dgeoId	This parameter must be set to **M_DEFAULT**.	c	e	f	g	a	i	j	k	l	m	n	b
PathOrTemplateIndex	Specifies the index of the template in the template 3D measurement context.	c	e	f	g	a	i	j	k	l	m	n	b
M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies the template to delete from the template 3D measurement context.	c	e	f	g	a	i	j	k	l	m	n	b
	Parameters												
	TemplateIndex Set this parameter to the following:												
	0 — Specifies the index of the template.	c	e	f	g	a	i	j	k	l	m	n	b

Return value

Reserved for future expansion.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasDraw

Synopsis

Draw specific features of templates, paths, profiles, or markers in an image buffer or 2D graphics list.

Syntax

```
void M3dmeasDraw(
    AIL_ID ContextGraId,            //in
    AIL_ID ContextOrResult3dmeasId, //in
    AIL_ID DstImageBufOrListGraId,  //out
    AIL_INT64 Operation,            //in
    AIL_INT64 PathOrTemplateIndex,  //in
    AIL_INT64 ProfileIndex,         //in
    AIL_INT64 MarkerIndex,          //in
    AIL_INT64 ControlFlag           //in
)
```

Description

This function draws specific template, path, profile, or marker features in the destination image buffer or 2D graphics list.

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context to use when drawing. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ContextOrResult3dmeasId

Specifies the 3D measurement context or result buffer from which to extract the features to draw. The 3D measurement context or result buffer must have been previously allocated using M3dmeasAlloc() or M3dmeasAllocResult(), respectively.

DstImageBufOrListGraId

Specifies the identifier of the destination image buffer or 2D graphics list in which to draw. The buffer can be any valid 1- or 3-band image buffer allocated using MbufAlloc...(). The 2D graphics list must be previously allocated using MgraAllocList(). By drawing into a display's overlay buffer or associating the 2D graphics list with the display, you can also annotate an image non-destructively.

If specifying an image buffer, it must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

Operation

Specifies the type of operation to perform. Operations can be added together to draw multiple features at a time. For example,

to draw both the marker's profile region and the marker, you would set the Operation parameter to M_DRAW_PROFILE_REGION + M_DRAW_MARKERS. The Operation parameter can be set to a combination of the following values.

The following operation can draw path features extracted from a path 3D measurement context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For extracting path features from a path 3D measurement context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_PATH	Draws the path as a line with an arrow head.	c		e	f	g	a		i		j	k	l	m	n	b

The following operation can draw template features extracted from a template 3D measurement context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For extracting template features from a template 3D measurement context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_TEMPLATE	Draws the template as a line with an arrow head.	c		e	f	g	a		i		j	k	l	m	n	b

The following operations can draw profile features extracted from a path or template 3D measurement context, or a profile, path, or template 3D measurement result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For extracting profile features from a path or template 3D measurement context or any 3D measurement result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_PROFILE_REGION	Draws a rectangle around the profile.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PROFILE_REGION_DIRECTION	Draws a line with an arrow pointing in the direction of the profile.	c		e	f	g	a		i		j	k	l	m	n	b

The following operations can draw marker features extracted from a profile, path, or template 3D measurement result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For extracting marker features from a 3D measurement result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_MARKERS	Draws a cross-like symbol at the marker position and, for pair markers, draws an H-type line (|-|) connecting the two transitions. Note that this is equivalent to M_DRAW_MARKERS_POSITION and M_DRAW_TRANSITIONS for single markers and M_DRAW_MARKERS_POSITION + M_DRAW_MARKERS_WIDTH for pair markers.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MARKERS_POSITION	Draws a cross-like symbol at the marker position. Note that this is equivalent to M_DRAW_MARKERS and M_DRAW_TRANSITIONS for single markers.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MARKERS_WIDTH	Draws an H-type line (|-|) connecting the two transitions to indicate the width of the pair marker. This operation is only supported for pair markers.	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_TRANSITIONS	Draws a cross-like symbol at the marker's transition positions. Note that this is equivalent to M_DRAW_MARKERS and M_DRAW_MARKERS_POSITION for single markers.	c		e	f	g	a		i		j	k	l	m	n	b

The following operations can draw fit features extracted from a template 3D measurement result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For extracting fit features from a template 3D measurement result buffer																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_FIT_GEOMETRY	Draws the fitted 3D line geometry.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIT_MARKERS	Draws a cross-like symbol at the marker positions that were used for the fit.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIT_MARKERS_OUTLIERS	Draws a cross-like symbol at the marker positions that were not used for the fit.	c		e	f	g	a		i		j	k	l	m	n	b

PathOrTemplateIndex

Specifies the index of the path or template in the specified 3D measurement context or result buffer.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the template or path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_PATH_INDEX(AIL_INT PathIndex)	Specifies that the drawing operation is for a path in the path 3D measurement context or result buffer, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PathIndex Set this parameter to the following:															
	0 — Specifies the index of the path.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies that the drawing operation is for a template in the template 3D measurement context or result buffer, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	TemplateIndex Set this parameter to the following:															

	0 Specifies the index of the template.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_TEMPLATE	Specifies that the drawing operation is for the default template in the profile 3D measurement result buffer, if one is specified. Note that the default template in a profile 3D measurement context is different from an explicitly defined template in a template 3D measurement context. The default template is inherent to the supplied profiles, such that it is the theoretical template that would result in these profiles. Note, unlike for the profiles perpendicular to the template in a template 3D measurement context, multiple markers can be found along the profiles perpendicular to the default template in a profile 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b

ProfileIndex

Specifies the index of the profile in the specified 3D measurement context or result buffer.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the profile																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_INDEX(AIL_INT Index)	Specifies that the drawing operation is for a profile in the specified 3D measurement context or result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Set this parameter to the following:															
	M_ALL Specifies all profiles in the specified 3D measurement context or result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

			Value >= 0	Specifies the index of the profile in the specified 3D measurement context or result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

MarkerIndex

Specifies the index of the marker in the specified 3D measurement result buffer.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the marker																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies that the drawing operation is for all markers in the specified 3D measurement result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the index of the marker in the specified 3D measurement result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion. this parameter must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasDraw3d

Synopsis

Draw specific features of templates, paths, profiles, or markers into a 3D graphics list.

Syntax

```
AIL_INT64 M3dmeasDraw3d(
    AIL_ID OperationDraw3dContext3dmeasId, //in
    AIL_ID SrcResult3dmeasId,              //in
    AIL_INT64 PathOrTemplateIndex,         //in
    AIL_INT64 ProfileIndex,                //in
    AIL_INT64 MarkerIndex,                 //in
    AIL_ID DstList3dgraId,                 //out
    AIL_INT64 DstParentLabel,              //in
    AIL_INT64 ControlFlag                  //in
)
```

Description

This function draws specific template, path, profile, or marker features, stored in the specified 3D measurement result buffer, into a 3D graphics list. Set the draw operations and options for the draw using M3dmeasControlDraw().

Parameters

OperationDraw3dContext3dmeasId

Specifies the identifier of the draw 3D measurement context that specifies the annotations to draw and how to draw them.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the draw 3D measurement context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies a predefined draw 3D measurement context. The default context is predefined with all draw operations (M3dmeasControlDraw()) set to use default settings.	c		e	f	g	a		i		j	k	l	m	n	b
Draw 3D measurement context identifier	Specifies a valid draw 3D measurement context identifier, which you have allocated using M3dmeasAlloc() with M_DRAW_3D_PATH_CONTEXT, M_DRAW_3D_PROFILE_CONTEXT, or M_DRAW_3D_TEMPLATE_CONTEXT.	c		e	f	g	a		i		j	k	l	m	n	b

SrcResult3dmeasId

Specifies the identifier of the profile, path, or template 3D measurement result buffer, previously allocated using M3dmeasAllocResult() with M_FIND_MARKER_PROFILE_RESULT, M_FIND_MARKER_PATH_RESULT, or M_FIND_MARKER_TEMPLATE_RESULT, respectively.

PathOrTemplateIndex

Specifies the path or template to draw. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a template or path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_PATH_INDEX(AIL_INT PathIndex)	Specifies the path in the path 3D measurement result buffer to draw, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PathIndex Set this parameter to the following:															
	0 — Specifies the index of the path.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies the template in the template 3D measurement result buffer to draw, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	TemplateIndex Set this parameter to the following:															
	0 — Specifies the index of the template.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_TEMPLATE	Specifies to draw the default template in the profile 3D measurement result buffer, if one is specified. Note that the default template in a profile 3D measurement context is different from an explicitly defined template in a template 3D measurement context. The default template is inherent to the supplied profiles, such that it is the theoretical template that would result in these profiles. Note, unlike for the profiles perpendicular to the template in a template 3D measurement context, multiple markers can be found along the profiles perpendicular to the default template in a profile 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b

ProfileIndex

Specifies the profile to draw. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a profile															
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_INDEX(AIL_INT Index)	Specifies the profile in the specified 3D measurement result buffer to draw.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	Index Set this parameter to the following:															

			M_ALL	Specifies all profiles.	c		e	f	g	a		i		j	k	l	m	n	b
			Value >= 0	Specifies the index of the profile.	c		e	f	g	a		i		j	k	l	m	n	b

MarkerIndex

Specifies the marker to draw. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a marker																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to draw features of all the markers.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the index of the markers for which to draw features.	c		e	f	g	a		i		j	k	l	m	n	b

DstList3dgraId

Specifies the identifier of the 3D graphics list in which to draw. The 3D graphics list must have been previously allocated using M3dgraAlloc(), or you can specify the identifier of the 3D display's internal graphics list (inquired using M3ddislnquire() with M_3D_GRAPHIC_LIST_ID).

DstParentLabel

Specifies the label of the 3D graphic in the 3D graphics list to use as the annotation's parent.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the parent label																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ROOT_NODE.	c		e	f	g	a		i		j	k	l	m	n	b
M_ROOT_NODE	Specifies the top-most node of the 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the label of the 3D graphic in the 3D graphics list.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Return value

Returns the label of the 3D graphic added to the 3D graphics list.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasFindMarker

Synopsis

Search for markers in a depth map.

Syntax

```
void M3dmeasFindMarker(
    AIL_ID Context3dmeasId,          //in
    AIL_ID DepthMapOrContainerBufId, //in
    AIL_ID Result3dmeasId,           //out
    AIL_INT64 ControlFlag            //in
)
```

Description

This function searches for markers in the specified depth map and writes the results in the specified result buffer. You can retrieve results using M3dmeasGetResult(). Markers denote the location of transitions along a profile. Edge transitions go from a lower Z-value to a higher Z-value (or vice versa). Note that invalid transitions can also be identified, either in the middle of an invalid region, or where the data goes from valid to invalid (or vice versa).

To find markers along a specified path in the supplied depth map, you must use a path 3D measurement context. You must add a path to the path 3D measurement context, using M3dmeasDefine(). Setting a path is useful when you want to find an object but you don't know its exact location. You can set a path that crosses through possible locations to detect height changes along the line. In this case, **M3dmeasFindMarker**() extracts a single profile along the path and finds the markers in the profile.

If you want to verify if an object is at an expected location or has an expected shape, you can find markers perpendicular to a template. To set a template where you expect a height change to occur, you must use a template 3D measurement context. You must add a template that follows an expected edge of your object to the template 3D measurement context, using M3dmeasDefine(). In this case, **M3dmeasFindMarker**() extracts profiles at regular intervals perpendicular to the template and finds the best marker close to the template in each profile.

If your template runs along a column(s) of the depth map, you can use a profile 3D measurement context instead, in which case the profile extraction step is skipped and the find marker operation is performed on each row (profile) of the depth map; you don't explicitly define the template. If you supply the profiles, the 3D measurement module can find the specified number of markers along each profile.

If you use a template 3D measurement context, you can use M3dmeasFit() after calling **M3dmeasFindMarker**() to fit a geometry to the found markers to compare its position with the defined template. Alternatively, you can find the markers from the template's perpendicular profiles and perform the fit simultaneously, using M3dmeasFit() with a template 3D measurement context and a depth map. When the find and fit are performed simultaneously, all possible markers are considered and the ones that provide the best fit are kept.

Use M3dmeasControl() to configure profile extraction settings and marker search settings before calling this function.

Parameters

Context3dmeasId

Specifies the 3D measurement context. The context must have been allocated using M3dmeasAlloc() with M_FIND_MARKER_PROFILE_CONTEXT, M_FIND_MARKER_PATH_CONTEXT, or M_FIND_MARKER_TEMPLATE_CONTEXT.

DepthMapOrContainerBufId

Specifies the identifier of the depth map container or depth map image buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the depth map container or depth map image buffer identifier																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

Depth map container identifier	Specifies the identifier of a depth map container. The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must store data in a 3D-processable depth map format (that is, if you call MbufInquireContainer() with M_3D_PROCESSABLE_DEPTH_MAP, the function returns M_TRUE). The container must not have a region component. Using a container with a region component will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b
Depth map image buffer identifier	Specifies the identifier of a depth map image buffer. The image buffer must be an 8-bit, 16-bit, or 32-bit unsigned, 1-band buffer, and must contain a fully corrected depth map (that is, if you call McalInquire() with M_DEPTH_MAP, the function returns M_TRUE). The image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b

Result3dmeasId

Specifies the identifier of the 3D measurement result buffer. The result buffer must have been allocated using M3dmeasAllocResult() with M_FIND_MARKER_PROFILE_RESULT, M_FIND_MARKER_PATH_RESULT, or M_FIND_MARKER_TEMPLATE_RESULT, and must match the specified context.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasFit

Synopsis

Fits a 3D line geometry to markers found in a template's perpendicular profiles.

Syntax

```
void M3dmeasFit(
    AIL_ID FitContext3dmeasId,       //in
    AIL_ID TemplateContext3dmeasId,  //in
    AIL_ID DepthMapOrContainerBufId, //in
    AIL_ID TemplateResult3dmeasId,   //out
    AIL_INT64 ControlFlag            //in
)
```

Description

This function fits a 3D line geometry to markers found in a template's perpendicular profiles. You can find the markers before calling this function, using M3dmeasFindMarker(). This function can also find the markers and perform the fit simultaneously, in which case, all possible markers are considered and the ones that provide the best fit are kept.

If finding the markers before calling this function, pass the template 3D measurement result buffer containing the markers to which to fit the line geometry.

If finding the markers and performing the fit simultaneously, you must provide the template 3D measurement context and the depth map image buffer or container in which to search for markers.

You can use M3dmeasGetResult() with M_ANGULARITY, M_CENTER_DISTANCE, and M_DISTANCE_MAX to compare the resulting fitted geometry to the defined template. Note that these comparison results are also available when the markers were previously found (because the template 3D measurement result buffer also holds the original template).

Parameters

FitContext3dmeasId

Specifies the identifier of the fit 3D measurement context. The context must have been allocated using M3dmeasAlloc() with M_FIT_CONTEXT.

TemplateContext3dmeasId

Specifies the identifier of the template 3D measurement context, when searching for the markers using this function. If TemplateResult3dmeasId already contains found markers and you only want to perform the fit operation, set this parameter to **M_NULL**.

The context must have been allocated using M3dmeasAlloc() with M_FIND_MARKER_TEMPLATE_CONTEXT.

DepthMapOrContainerBufId

Specifies the identifier of the depth map container or depth map image buffer, when searching for the markers using this function. If TemplateResult3dmeasId already contains found markers and you only want to perform the fit operation, set this parameter to **M_NULL**.

The container must have been previously allocated using MbufAllocContainer() with M_PROC, and must store data in a 3D-processable depth map format (that is, if you call MbufInquireContainer() with M_3D_PROCESSABLE_DEPTH_MAP, the function returns M_TRUE).

The image buffer must be an 8-bit, 16-bit, or 32-bit unsigned, 1-band buffer, and must contain a fully corrected depth map (that is, if you call McalInquire() with M_DEPTH_MAP, the function returns M_TRUE).

TemplateResult3dmeasId

Specifies the identifier of the template 3D measurement result buffer. The result buffer must have been allocated using M3dmeasAllocResult() with M_FIND_MARKER_TEMPLATE_RESULT, and, if TemplateContext3dmeasId and DepthMapOrContainerBufId are set to **M_NULL**, must contain the results of a call to M3dmeasFindMarker().

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasFree

Synopsis

Free a 3D measurement context or result buffer.

Syntax

```
void M3dmeasFree(
    AIL_ID ContextOrResult3dmeasId //in
)
```

Description

This function deletes the specified 3D measurement context or result buffer identifier, and releases any memory associated with it.

All 3D measurement contexts and result buffers allocated on a particular system must be freed before the system can be freed, unless M_UNIQUE_ID was specified during allocation.

Parameter

ContextOrResult3dmeasId

Specifies the identifier of the 3D measurement context or result buffer to free. These must have been successfully allocated using M3dmeasAlloc() or M3dmeasAllocResult() prior to calling this function.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasGetResult

Synopsis

Get the specified type of result(s) from a 3D measurement result buffer.

Syntax

```
AIL_DOUBLE M3dmeasGetResult(
    AIL_ID Result3dmeasId,         //in
    AIL_INT64 PathOrTemplateIndex, //in
    AIL_INT64 ProfileIndex,        //in
    AIL_INT64 MarkerIndex,         //in
    AIL_INT64 ResultType,          //in
    void *ResultArrayPtr           //out
)
```

Description

This function retrieves the result(s) of the specified type from a 3D measurement result buffer.

Parameters

Result3dmeasId

Specifies the identifier of the 3D measurement result buffer from which to retrieve results.

PathOrTemplateIndex

Specifies the index of the path or template, or the default template in a profile 3D measurement result buffer. Set this parameter to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of the template or path

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_PATH_INDEX(AIL_INT PathIndex)	Specifies to retrieve results for a path in a path 3D measurement result buffer, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PathIndex Set this parameter to the following:															
	0 — Specifies the index of the path.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies to retrieve results for a template in a template 3D measurement result buffer, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	TemplateIndex Set this parameter to the following:															

	0 Specifies the index of the template.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_TEMPLATE	Specifies to retrieve results for the default template in a profile 3D measurement result buffer, if one is specified. Note that the default template in a profile 3D measurement context is different from an explicitly defined template in a template 3D measurement context. The default template is inherent to the supplied profiles, such that it is the theoretical template that would result in these profiles. Note, unlike for the profiles perpendicular to the template in a template 3D measurement context, multiple markers can be found along the profiles perpendicular to the default template in a profile 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to retrieve general results relating to the entire 3D measurement result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

ProfileIndex

Specifies the index of the profile. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of the profile																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default value. If the PathOrTemplateIndex parameter is set to M_DEFAULT or M_GENERAL, specifies that this parameter is unused; otherwise, same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_INDEX(AIL_INT ProfileIndex)	Specifies to retrieve results for a profile.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	ProfileIndex Set this parameter to one of the following:															
	M_ALL Specifies all profiles.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0 Specifies the index of the profile.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to retrieve general results for the default template, a path, or a template.	c		e	f	g	a		i		j	k	l	m	n	b

MarkerIndex

Specifies the index of the marker. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the index of the marker																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies the default value. If the ProfileIndex parameter is set to M_DEFAULT or M_GENERAL, specifies that this parameter is unused; otherwise, same as M_GENERAL.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSITION_FIRST(AIL_INT MarkerIndex)	Specifies to retrieve results for the first transition of a marker. Note that the first transition is the sole transition of an M_SINGLE marker.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	MarkerIndex Set this parameter to one of the following:															
	M_ALL — Specifies all markers.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* < M_NUMBER_MARKER — Specifies the index of the marker.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSITION_SECOND(AIL_INT MarkerIndex)	Specifies to retrieve results for the second transition of an M_PAIR marker.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	MarkerIndex Set this parameter to one of the following:															
	M_ALL — Specifies all markers.	c		e	f	g	a		i		j	k	l	m	n	b
	0 <= *Value* < M_NUMBER_MARKER — Specifies the index of the marker.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL	Specifies to retrieve results for all markers.	c		e	f	g	a		i		j	k	l	m	n	b
M_GENERAL	Specifies to retrieve general results for a profile.	c		e	f	g	a		i		j	k	l	m	n	b
0 <= *Value* < M_NUMBER_MARKER	Specifies the index of the individual marker for which to retrieve results.	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result to retrieve.

See the Parameter associations section for possible values that can be specified.

ResultArrayPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]
- array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]
- array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]
- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]
- AIL_DOUBLE
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the results. Since the **M3dmeasGetResult**() function also returns the results, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **ResultType** parameter and possible values returned to the **ResultArrayPtr** parameter.

- For retrieving general results from a profile, path, or template 3D measurement result buffer
- For retrieving default template, path, or template results from a profile, path, or template 3D measurement result buffer
- For retrieving profile results from a profile, path, or template 3D measurement result buffer
- For retrieving marker or transition results from a profile, path, or template 3D measurement result buffer
- For retrieving fit results from a template 3D measurement result buffer

When retrieving general results from a profile, path, or template 3D measurement result buffer, the ResultType parameter can be set to the following value. The PathOrTemplateIndex parameter must be set to M_GENERAL, and the ProfileIndex and MarkerIndex parameters must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving general results from a profile, path, or template 3D measurement result buffer																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																	
M_STATUS_FIND +	Retrieves the status of the find marker operation. This result is always available. For the status of a fit operation, use M_STATUS_FIT.																
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETE	Specifies that the find marker operation completed.	c		e	f	g	a		i		j	k	l	m	n	b
M_CURRENTLY_CALCULATING	Specifies that the find marker operation is ongoing. You can only get this status if you are retrieving it from another thread.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERNAL_ERROR	Specifies that an unexpected internal error occurred during the find marker operation.	c		e	f	g	a		i		j	k	l	m	n	b

	M_NOT_INITIALIZED	Specifies that the 3D measurement result buffer was not used in a call to M3dmeasFindMarker() or M3dmeasFit(), and contains no results.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STOPPED_BY_REQUEST	Specifies that the find marker operation was stopped from another thread using M3dmeasControl() with M_STOP_FIND.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TIMEOUT_REACHED	Specifies that the find marker operation took longer than the allowed value, specified using M3dmeasControl() with M_TIMEOUT, and has stopped before completion.	c		e	f	g	a		i		j	k	l	m	n	b

When retrieving results for a default template, path, or template from a profile, path, or template 3D measurement result buffer, the ResultType parameter can be set to one of the following values. The PathOrTemplateIndex parameter must be set to M_DEFAULT_TEMPLATE, M_PATH_INDEX(), or M_TEMPLATE_INDEX(), the ProfileIndex parameter must be set to M_GENERAL, and the MarkerIndex parameter must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving default template, path, or template results from a profile, path, or template 3D measurement result buffer																	
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ResultArrayPtr - Possible values returned																
M_NUMBER_PROFILE +		Retrieves the number of profiles that were analyzed. Note that along a path, only one profile is extracted and analyzed.															
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE +		Retrieves the projection angle used to create the profile. This is also the angle, relative to the profile line, at which markers are found. Note that this result type is not available for a profile 3D measurement result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_SAMPLE_SIZE +		Retrieves the sampling distance between profile points, in world units. Note that this is also the pixel size in X of the profiles. For a path or template 3D measurement result buffer, this is equivalent to the sampling distance specified using M3dmeasControl() with M_PROFILE_SAMPLE_SIZE when M_PROFILE_SAMPLE_SIZE_MODE is set to M_ABSOLUTE, or equivalent to M_PROFILE_SAMPLE_SIZE multiplied by the source pixel size when M_PROFILE_SAMPLE_SIZE_MODE is set to M_RELATIVE_TO_PIXEL_SIZE_....	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

When retrieving results for a profile from a profile, path, or template 3D measurement result buffer, the ResultType parameter can be set to one of the following values. The PathOrTemplateIndex parameter must be set to M_DEFAULT_TEMPLATE, M_PATH_INDEX(), or M_TEMPLATE_INDEX(), the ProfileIndex parameter must be set to the index of one or all profiles, and the MarkerIndex parameter must be set to M_GENERAL.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving profile results from a profile, path, or template 3D measurement result buffer

ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_NUMBER_MARKER +	Retrieves the number of markers found. Note that for a template 3D measurement result buffer, a maximum of 1 marker is found for each profile taken perpendicular to a template.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_RESPONSE +	Retrieves the response of the profile after applying a filter (that is, the normalized first derivative edge profile response or normalized second derivative ridge profile response). Note that this result type is not available if the M_MARKER_TRANSITION control type is set to M_INVALID.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT_SIZE_X +	Retrieves the size in X of the profile response, in sample units. Note that this is equivalent to the number of profile points.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b

When retrieving results for a marker or marker transition from a profile, path, or template 3D measurement result buffer, the ResultType parameter can be set to one of the following values. The PathOrTemplateIndex parameter must be set to M_DEFAULT_TEMPLATE, M_PATH_INDEX(), or M_TEMPLATE_INDEX(), the ProfileIndex parameter must be set to the index of one or all profiles, and the MarkerIndex parameter must be set to the index of one or all markers or the first or second transition of one or all markers.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving marker or transition results from a profile, path, or template 3D measurement result buffer

ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_DEPTH +	Retrieves the depth of the marker or transition. This is equivalent to M_MAX_Z - M_MIN_Z.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_INLIER_STATUS +	Retrieves whether the marker was inside or outside the fit. Note that the MarkerIndex parameter must be set to the index of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_INSIDE	Specifies that the marker was inside.	c		e	f	g	a		i		j	k	l	m	n	b
M_OUTSIDE	Specifies that the marker was outside.	c		e	f	g	a		i		j	k	l	m	n	b

M_MARKER_POSITION_END +	Retrieves the position of the end of the marker relative to the start of the profile, in world units. The end position of an M_SINGLE marker is the end of the transition. The end position of an M_PAIR marker is the center of the second transition. Note that the MarkerIndex parameter must be set to the index of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_POSITION_START +	Retrieves the position of the start of the marker relative to the start of the profile, in world units. The start position of an M_SINGLE marker is the start of the transition. The start position of an M_PAIR marker is the center of the first transition. Note that the MarkerIndex parameter must be set to the index of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_Z +	Retrieves the maximum Z-value that makes up the marker or transition.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_Z +	Retrieves the minimum Z-value that makes up the marker or transition.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_OFFSET +	Retrieves the offset of the marker or transition from the position specified using M3dmeasControl() with M_MARKER_POSITION_RELATIVE. The offset is calculated using the following formula: (M_POSITION / *nominal profile length* * 100) - M_MARKER_POSITION_RELATIVE . For example, if the marker/transition was found at 3 mm from the start of the profile, the nominal profile length is 10 mm, and M_MARKER_POSITION_RELATIVE is set to 50%, M_OFFSET will return -20%. Use M_POSITION to retrieve the position relative to the start of the profile.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POLARITY +	Retrieves the polarity of the transition. Note that the MarkerIndex parameter must be set to the first or second transition of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b

	M_NEGATIVE	Specifies a negative polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITIVE	Specifies a positive polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_UNDEFINED	Specifies an undefined polarity because the transition is an M_INVALID transition with no depth.	c		e	f	g	a		i		j	k	l	m	n	b
M_POLARITY_PAIR +		Retrieves the polarity of the M_PAIR marker. Note that the MarkerIndex parameter must be set to the index of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPPOSITE	Specifies that the polarity of the second transition is the opposite of that of the first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the polarity of the two transitions are the same.	c		e	f	g	a		i		j	k	l	m	n	b
	M_UNDEFINED	Specifies an undefined polarity because one or both of the transitions is an M_INVALID transition with no depth.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION +		Retrieves the position of the marker or transition relative to the start of the profile, in world units. For example, if the marker/transition was found at 3 mm from the start of the profile, M_POSITION will return 3 mm. Use M_OFFSET to retrieve the position relative to another profile position.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_EXTRACTED +		Retrieves the position of the marker or transition relative to the start of the profile, in sample units. Note that this considers the entire profile, including areas that are clipped due to falling outside of the depth map. The extracted position is calculated using the following formula: M_POSITION / (M_PROFILE_SAMPLE_SIZE / sample). For example, if the marker/transition was found at 3 mm from the start of the profile and the sampling distance is 0.25 mm, M_POSITION_EXTRACTED will return 12 sample units.	c		e	f	g	a		i		j	k	l	m	n	b
	ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_EXTRACTED_CLIPPED +		Retrieves the position of the marker or transition relative to the start of the clipped profile, in sample units. For example, if the marker or transition is 12 sample units from the start of whole profile, but the first 2 sample units of the profile fall outside of the depth map, M_POSITION_EXTRACTED_CLIPPED will return 10 sample units.	c		e	f	g	a		i		j	k	l	m	n	b

ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_INVALID +	Retrieves the position of the invalid transition relative to an invalid region (gap) in the profile. You can specify where to identify a transition when an invalid region is encountered in the profile, using M3dmeasControl() with M_MARKER_POSITION_INVALID. Note that the MarkerIndex parameter must be set to the first or second transition of one or all markers. This result type is only meaningful if the M_TRANSITION result type is M_INVALID.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_AFTER	Specifies that the transition is located after the gap.	c		e	f	g	a		i		j	k	l	m	n	b
M_BEFORE	Specifies that the transition is located before the gap.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIDDLE	Specifies that the transition is located in the middle of the gap.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_RELATIVE +	Retrieves the position of the marker or transition relative to the start of the profile, as a percentage of the nominal length of the profile. The relative position is calculated using the following formula: M_POSITION / *nominal profile length* * 100. For example, if the marker/transition was found at 3 mm from the start of the profile and the nominal profile length is 10 mm, M_POSITION_RELATIVE will return 30%.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_X +	Retrieves the X-coordinate of the marker or transition in the source depth map, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y +	Retrieves the Y-coordinate of the marker or transition in the source depth map, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Z +	Retrieves the Z-coordinate of the marker or transition in the source depth map, in world units.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_SPACING +	Retrieves the spacing between the marker and the next marker. The spacing is equivalent to M_POSITION of the next marker minus M_POSITION of the current marker. Note that the spacing of the last marker is always 0. Note that the MarkerIndex parameter must be set to the index of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_SPACING_RELATIVE +	Retrieves the spacing between the marker and the next marker, as a percentage of the nominal length of the profile. The relative spacing is calculated using the following formula: M_SPACING / *nominal profile length* * 100. For example, if the spacing is 4mm and the nominal profile length is 10 mm, M_SPACING_RELATIVE will return 40%. Note that the MarkerIndex parameter must be set to the index of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_STRENGTH +	Retrieves the strength of the marker or transition in the profile response.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSITION +	Retrieves the type of transition. Note that the MarkerIndex parameter must be set to the first or second transition of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT* **|** *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_EDGE	Specifies an edge transition.	c		e	f	g	a		i		j	k	l	m	n	b
M_INVALID	Specifies an invalid transition. If M3dmeasControl() with M_MARKER_TRANSITION is set to M_EDGE_OR_INVALID or M_INVALID, invalid transitions can be identified before, after, or in the middle of gaps encountered in the profile.	c		e	f	g	a		i		j	k	l	m	n	b
M_RIDGE	Specifies a ridge transition.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSITION_POSITION_END +	Retrieves the position of the end of the transition relative to the start of the profile, in world units. Note that the MarkerIndex parameter must be set to the first or second transition of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

M_TRANSITION_POSITION_START +	Retrieves the position of the start of the transition relative to the start of the profile, in world units. Note that the MarkerIndex parameter must be set to the first or second transition of one or all markers.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_WIDTH +	Retrieves the width of the marker or transition. The width of a transition or an M_SINGLE marker is the distance between the start and end of the transition. The width of an M_PAIR marker is the distance between the center of the first transition and center of the second transition. Note that for a transition, the width is equivalent to M_TRANSITION_POSITION_START - M_TRANSITION_POSITION_END. For a marker, the width is equivalent to M_MARKER_POSITION_START - M_MARKER_POSITION_END.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
M_WIDTH_RELATIVE +	Retrieves the width of the marker or transition, as a percentage of the nominal length of the profile. The relative width is calculated using the following formula: M_WIDTH / *nominal profile length* * 100. For example, if the marker/transition is 2 mm wide and the nominal profile length is 10 mm, M_WIDTH_RELATIVE will return 20%.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

When retrieving fit results for a template from a template 3D measurement result buffer, the ResultType parameter can be set to one of the following values. The PathOrTemplateIndex parameter must be set to M_TEMPLATE_INDEX(), the ProfileIndex parameter must be set to M_GENERAL, and the MarkerIndex parameter must be set to M_DEFAULT. These results are only available after a call to M3dmeasFit(), unless otherwise specified.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving fit results from a template 3D measurement result buffer																
ResultType ResultArrayPtr - Possible values returned	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_ANGULARITY +	Retrieves the angle between the fitted 3D geometry and the template, in degrees.															
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_AXIS_X +	Retrieves the X-component of the fitted 3D geometry's unit vector. For a 3D line geometry, the X-component of the line's direction unit vector is retrieved. This vector does not reflect the line's length.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

M_AXIS_Y +	Retrieves the Y-component of the fitted 3D geometry's unit vector. For a 3D line geometry, the Y-component of the line's direction unit vector is retrieved. This vector does not reflect the line's length.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_AXIS_Z +	Retrieves the Z-component of the fitted 3D geometry's unit vector. For a 3D line geometry, the Z-component of the line's direction unit vector is retrieved. This vector does not reflect the line's length.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_DISTANCE +	Retrieves the distance between the center of the fitted 3D geometry and the template.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_X +	Retrieves the X-coordinate of the center of the fitted 3D geometry.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Y +	Retrieves the Y-coordinate of the center of the fitted 3D geometry.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_Z +	Retrieves the Z-coordinate of the center of the fitted 3D geometry.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISTANCE_MAX +	Retrieves the maximum distance between the fitted 3D geometry and the template.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_END_POINT_X +	Retrieves the X-coordinate of the fitted 3D geometry's end point.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_END_POINT_Y +	Retrieves the Y-coordinate of the fitted 3D geometry's end point.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_END_POINT_Z +	Retrieves the Z-coordinate of the fitted 3D geometry's end point.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_RMS_ERROR +	Retrieves the root-mean-squared (RMS) error of the distance between the markers and the fitted 3D geometry. Only inliers are considered when calculating the RMS error.	c		e	f	g	a		i		j	k	l	m	n	b

ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_MARKER_FITTED +	Retrieves the number of markers that were used for the fit operation.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_MARKER_OUTLIERS +	Retrieves the number of markers that were not used for the fit operation.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_START_POINT_X +	Retrieves the X-coordinate of the fitted 3D geometry's start point.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_START_POINT_Y +	Retrieves the Y-coordinate of the fitted 3D geometry's start point.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_START_POINT_Z +	Retrieves the Z-coordinate of the fitted 3D geometry's start point.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_STATUS_FIT +	Retrieves the status of the fit operation. This result is always available. For the status of a find marker operation, use M_STATUS_FIND.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_COMPLETE	Specifies that the fit operation completed.	c		e	f	g	a		i		j	k	l	m	n	b
M_CURRENTLY_CALCULATING	Specifies that the fit operation is ongoing. You can only get this status if you are retrieving it from another thread.	c		e	f	g	a		i		j	k	l	m	n	b
M_INSUFFICIENT_COVERAGE	Specifies that the fit operation failed because an insufficient number of markers were covered by the fitted points.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERNAL_ERROR	Specifies that an unexpected error occurred during the fit operation.	c		e	f	g	a		i		j	k	l	m	n	b
M_NOT_ENOUGH_VALID_DATA	Specifies that the fit operation failed because there were not enough valid points to fit the specified 3D geometry.	c		e	f	g	a		i		j	k	l	m	n	b
M_NOT_INITIALIZED	Specifies that the 3D measurement result buffer was not used in a call to M3dmeasFit(), and contains no results.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for

Tables:

- For retrieving general results from a profile, path, or template 3D measurement result buffer
- For retrieving default template, path, or template results from a profile, path, or template 3D measurement result buffer
- For retrieving profile results from a profile, path, or template 3D measurement result buffer

- For retrieving marker or transition results from a profile, path, or template 3D measurement result buffer
- For retrieving fit results from a template 3D measurement result buffer

the values listed in all the parameter association tables.

You can add the following value to the above-mentioned values to specify whether a result is available.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining whether results are available																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																	
M_AVAILABLE	Retrieves whether the requested result type is available for retrieval.																
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the requested result type is not available.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the requested result type is available.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values listed in For retrieving profile results from a profile, path, or template 3D measurement result buffer, For retrieving marker or transition results from a profile, path, or template 3D measurement result buffer.

You can add the following value to the above-mentioned values to determine the required array size (number of elements) to store the returned values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining the required array size (number of elements) to store the returned values																	
ResultType combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																	
M_NB_ELEMENTS	Retrieves the required array size (number of elements) to store the returned values.																
ResultArrayPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For retrieving general results from a profile, path, or template 3D measurement result buffer
- For retrieving default template, path, or template results from a profile, path, or template 3D measurement result buffer
- For retrieving profile results from a profile, path, or template 3D measurement result buffer
- For retrieving marker or transition results from a profile, path, or template 3D measurement result buffer
- For retrieving fit results from a template 3D measurement result buffer

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested results to the required data type.

> Note: In C++, when using a standard vector (std::vector) instead of an array, the values listed in this combination table must not be used. If you need to use a specific data type, instantiate the vector with the necessary data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																		
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
ResultArrayPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested results to an *AIL_DOUBLE*.															
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* **|** *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested results to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]* **|** *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested results to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]* **|** *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested results to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
ResultArrayPtr extra info	Data type info: Data type: *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]* **|** *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_DOUBLE*. If the requested information does not fit into an *AIL_DOUBLE*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasInquire

Synopsis

Inquire information about a 3D measurement context.

Syntax

```
AIL_INT64 M3dmeasInquire(
    AIL_ID Context3dmeasId,        //in
    AIL_INT64 PathOrTemplateIndex, //in
    AIL_INT64 ProfileIndex,        //in
    AIL_INT64 InquireType,         //in
    void *UserVarPtr               //out
)
```

Description

This function inquires information about a 3D measurement context.

To retrieve results from a 3D measurement result buffer, use M3dmeasGetResult().

To inquire about draw 3D measurement settings for M3dmeasDraw3d(), use M3dmeasInquireDraw() instead.

Parameters

Context3dmeasId

Specifies the identifier of the 3D measurement context to inquire. The 3D measurement context must have been previously allocated on the required system using M3dmeasAlloc().

PathOrTemplateIndex

Specifies that information will be inquired about a 3D measurement context, a path, a template, or a default template. Set this parameter to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying what to inquire																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_CONTEXT.	c		e	f	g	a		i		j	k	l	m	n	b
M_PATH_INDEX(AIL_INT PathIndex)	Specifies to inquire about an individual path in a path 3D measurement context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	PathIndex Set this parameter to the following:															
	0 — Specifies the index of the path.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_INDEX(AIL_INT TemplateIndex)	Specifies to inquire about an individual template in a template 3D measurement context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															

	TemplateIndex Set this parameter to the following:															
	0 — Specifies the index of the template.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTEXT	Specifies to inquire about a global setting of a 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_TEMPLATE	Specifies to inquire about the default template in a profile 3D measurement context, if one is specified. Note that the default template in a profile 3D measurement context is different from an explicitly defined template in a template 3D measurement context. The default template is inherent to the supplied profiles, such that it is the theoretical template that would result in these profiles. Note, unlike for the profiles perpendicular to the template in a template 3D measurement context, multiple markers can be found along the profiles perpendicular to the default template in a profile 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b

ProfileIndex

Specifies the profile to inquire about, if required. Set this parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a profile																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the index of a profile is not required.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_INDEX(AIL_INT ProfileIndex)	Specifies to inquire about an individual profile in a template 3D measurement context, if one is specified.	c		e	f	g	a		i		j	k	l	m	n	b
	Parameters															
	ProfileIndex Set this parameter to one of the following:															
	M_ALL — Specifies all profiles.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0 — Specifies the index of the profile.	c		e	f	g	a		i		j	k	l	m	n	b

InquireType

Specifies the type of information about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_FLOAT
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address of the variable in which to write the requested information. Since the **M3dmeasInquire**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr** parameter.

- For inquiring about a global setting of a profile, path, or template 3D measurement context
- For inquiring about the default template in a profile 3D measurement context, path in a path 3D measurement context, or template in a template 3D measurement context
- For inquiring about the profile extraction along a path in a path 3D measurement context or perpendicular to the template in a template 3D measurement context
- For inquiring about a template in a template 3D measurement context
- For inquiring about a template or profile in a template 3D measurement context
- For inquiring about a profile in a template 3D measurement context
- For inquiring about a fit 3D measurement context
- For inquiring about a path 3D measurement context
- For inquiring about a template 3D measurement context

For a profile, path, or template 3D measurement context, the InquireType parameter can be set to one of the following to inquire about a global setting. PathOrTemplateIndex must be set to M_DEFAULT or M_CONTEXT, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a global setting of a profile, path, or template 3D measurement context																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_TIMEOUT +		Inquires the maximum amount of time for M3dmeasFindMarker() or M3dmeasFit() to complete the find marker operation before generating a time-out error. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that there is no timeout value.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the timeout value, in msec.	c		e	f	g	a		i		j	k	l	m	n	b

For the default template in a profile 3D measurement context, a path in a path 3D measurement context, or a template in a template 3D measurement context, the InquireType parameter can be set to one of the following. PathOrTemplateIndex must be set to M_DEFAULT_TEMPLATE, the index of a path, or the index of a template, respectively, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the default template in a profile 3D measurement context, path in a path 3D measurement context, or template in a template 3D measurement context

InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_MARKER_DEPTH_MIN +		Inquires the minimum depth required for a peak or valley in the profile response, for it to be considered a valid transition signifying a marker. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the minimum depth. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_NUMBER +		Inquires the number of markers to search for in the profile. Note that this inquire type is not available for a template 3D measurement context. In this case, the find marker operation always looks for 1 marker in each perpendicular profile of the template. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL	Specifies to search for all markers.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of markers for which to search.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_NUMBER_TRANSITION_INSIDE +		Inquires the number of transitions to find inside a marker. This inquire type is only available when M_MARKER_TYPE is set to M_PAIR. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies to find markers with any number of transitions.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the number of transitions.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_POLARITY +		Inquires the polarity of the transition to find. This is the polarity of the sole transition for an M_SINGLE marker and the first transition for an M_PAIR marker. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies to find a transition with any polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NEGATIVE	Specifies to find a transition with negative polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITIVE	Specifies to find a transition with positive polarity.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_POLARITY_SECOND +		Inquires the polarity of the second transition to find. This is the polarity of the second transition for an M_PAIR marker. This inquire type is only available when M_MARKER_TYPE is set to M_PAIR. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies to find a transition with any polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NEGATIVE	Specifies to find a transition with negative polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPPOSITE	Specifies to find a transition with the opposite polarity of the first transition.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITIVE	Specifies to find a transition with positive polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies to find a transition with the same polarity as the first transition.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_POSITION_INVALID +		Inquires where to identify a transition when an invalid region is encountered in the profile. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AFTER	Specifies that the transition is immediately after the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BEFORE	Specifies that the transition is immediately before the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOTH	Specifies that there is one transition immediately before and one transition immediately after the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIDDLE	Specifies that the transition is in the middle of the gap.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_POSITION_RELATIVE +		Inquires the position that the markers should be closest to when M_MARKER_SELECTION is set to M_POSITION_RELATIVE and the number of candidates is greater than M_MARKER_NUMBER. Note that this also inquires the relative position used when calculating the M3dmeasGetResult() M_OFFSET result type. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the position, as a percentage of the nominal profile length.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_SELECTION +		Inquires the policy used to select found markers when the number of candidates is greater than M_MARKER_NUMBER. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_BEST_FIT	Specifies to use a different selection criteria depending on the operation. When performing an M3dmeasFindMarker() operation, M_BEST_FIT specifies to select markers according to their proximity to the profile's origin; the closest marker is selected first. When performing an M3dmeasFit() operation with a template 3D measurement context, M_BEST_FIT specifies to select markers according to possible resulting fit scores; the marker that would result in the best fit possible is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSEST_TO_SPACING	Specifies to select markers according to their similarity to the spacing set using M_MARKER_SPACING_RELATIVE; the marker with the closest spacing is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSEST_TO_WIDTH	Specifies to select markers according to their similarity to the width set using M_MARKER_WIDTH_RELATIVE; the marker with the closest width is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NARROWEST	Specifies to select markers according to their width; the narrowest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_FIRST	Specifies to select markers according to their proximity to the profile's origin; the closest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_LAST	Specifies to select markers according to their proximity to the end of the profile; the closest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_RELATIVE	Specifies to select markers according to their proximity to the position set using M_MARKER_POSITION_RELATIVE; the closest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STRONGEST	Specifies to select markers according to the strength of the transition; the marker with the strongest transition is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TALLEST	Specifies to select markers according to their measured depth; the marker with the greatest depth is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WIDEST	Specifies to select markers according to their width; the widest marker is selected first.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_SPACING_RELATIVE +		Inquires the spacing that the marker's spacing should be closest to when M_MARKER_SELECTION is set to M_CLOSEST_TO_SPACING and the number of candidates is greater than M_MARKER_NUMBER. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the spacing, as a percentage of the nominal profile length.	c		e	f	g	a		i		j	k	l	m	n	b

M_MARKER_STRENGTH_MIN +		Inquires the minimum strength required for a peak or valley in the profile response, for it to be considered a valid transition signifying a marker. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the minimum strength. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_STRENGTH_MIN_VAR +		Inquires the minimum prominence required for a peak or valley in the profile response, for it to be considered a valid transition signifying a marker. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the minimum prominence. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_TRANSITION +		Inquires the type of transition to find. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGE	Specifies to find edge transitions.	c		e	f	g	a		i		j	k	l	m	n	b
	M_EDGE_OR_INVALID	Specifies to find edge transitions and invalid transitions. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_INVALID	Specifies to find invalid transitions. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RIDGE	Specifies to find ridge transitions.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_TYPE +		Inquires the type of marker to extract from the profile. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_PAIR	Specifies to extract markers represented by a pair of transitions.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SINGLE	Specifies to extract markers represented by a single transition.	c		e	f	g	a		i		j	k	l	m	n	b
M_MARKER_WIDTH_RELATIVE +		Inquires the width that the marker's width should be closest to when M_MARKER_SELECTION is set to M_CLOSEST_TO_WIDTH and the number of candidates is greater than M_MARKER_NUMBER. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the width, as a percentage of the nominal profile length.	c		e	f	g	a		i		j	k	l	m	n	b

M_PROFILE_FILTER_SMOOTHNESS +		Inquires the normalized degree of smoothness (strength of denoising) applied to the profile to create the profile response before finding the markers. This inquire type is only used when M_PROFILE_FILTER_SMOOTHNESS_TYPE is set to M_NORMALIZED. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the smoothness value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_FILTER_SMOOTHNESS_SIZE +		Inquires the smoothness size applied to the profile to create the profile response before finding the markers. This inquire type is only used when M_PROFILE_FILTER_SMOOTHNESS_TYPE is set to M_SIZE. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 2.0	Specifies the smoothness size.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_FILTER_SMOOTHNESS_TYPE +		Inquires the filter smoothness type. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORMALIZED	Specifies to apply a normalized filter smoothness, set with M_PROFILE_FILTER_SMOOTHNESS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SIZE	Specifies to apply a filter smoothness size, set with M_PROFILE_FILTER_SMOOTHNESS_SIZE.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_GAP_FILL_SHARP_ELEVATION +		Inquires how to fill sharp elevation gaps. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AFTER	Specifies to fill the gap with the value of the boundary after the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BEFORE	Specifies to fill the gap with the value of the boundary before the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to fill sharp elevation gaps.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX	Specifies to fill the gap with the maximum gap boundary value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIDDLE	Specifies to fill the first half with the value of the boundary before the gap and the second half with the value of the boundary after the gap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN	Specifies to fill the gap with the minimum gap boundary value.	c		e	f	g	a		i		j	k	l	m	n	b

M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH +		Inquires the threshold used to differentiate between gradual elevation gaps and sharp elevation gaps. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies that the threshold is infinite. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the threshold, as a percentage of the depth map's Z-range or in world units, depending on the units specified with M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH_UNITS. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH_UNITS +		Inquires the units with which to interpret the M_PROFILE_GAP_FILL_SHARP_ELEVATION_DEPTH inquire type. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_PERCENTAGE	Specifies to interpret the value as a percentage of the range of Z-values (Z-range) visible in the depth map.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD	Specifies to interpret the value in world units.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_GAP_FILL_THRESHOLD +		Inquires the maximum size of gaps that will be filled. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no maximum gap size.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the maximum gap size, in sample or world units, depending on the units specified with M_PROFILE_GAP_FILL_THRESHOLD_UNITS.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_GAP_FILL_THRESHOLD_UNITS +		Inquires the units with which to interpret the M_PROFILE_GAP_FILL_THRESHOLD inquire type. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAMPLE	Specifies to interpret the value relative to the sampling distance (M_PROFILE_SAMPLE_SIZE).	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD	Specifies to interpret the value in world units.	c		e	f	g	a		i		j	k	l	m	n	b

For a path in a path 3D measurement context or template in a template 3D measurement context, the InquireType parameter can be set to one of the following. PathOrTemplateIndex must be set to the index of a path or template, respectively, and ProfileIndex must be set to M_DEFAULT. These inquire types are used to inquire about how to extract the profile along the path or the profiles at regular intervals perpendicular to the template.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the profile extraction along a path in a path 3D measurement context or perpendicular to the template in a template 3D measurement context																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

	UserVarPtr - Possible values returned															
M_PROFILE_ACCUMULATE_TYPE +		Inquires how points in a given sample are accumulated to create a single profile point when there are multiple points within the thickness of the profile line. SET														
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_MAX	Specifies to use the largest value of all the points in the sample. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEAN	Specifies to use the mean value of all the points in the sample.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIN	Specifies to use the smallest value of all the points in the sample. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_INTERPOLATION_AVERAGE_FRACTION +		Inquires the fraction of the sampling distance to use when taking a weighted average of the area surrounding the profile point. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 1.0	Specifies the fraction.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_INTERPOLATION_MODE +		Inquires the interpolation mode used to calculate the value for each point in the sample that will be used to create the profile point. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_AVERAGE	Specifies averaging interpolation. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_BILINEAR	Specifies bilinear interpolation. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_NEAREST_NEIGHBOR	Specifies nearest neighbor interpolation. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_MIN_VALID_PERCENTAGE +		Inquires the minimum percentage of points in a given sample that must be valid to set the corresponding profile point to a valid value. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the percentage of points in the sample. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE +		Inquires the nominal angle at which the projection is done for a given profile. SET	c		e	f	g	a		i		j	k	l	m	n	b

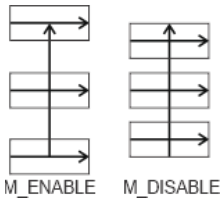
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	10.0 <= *Value* <= 170.0	Specifies the nominal projection angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE_ACCURACY +		Inquires the accuracy of the projection angle. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	10.0 <= *Value* <= 170.0	Specifies the nominal projection angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE_DELTA_NEG +		Inquires the lower limit of the projection angle range, relative to the nominal projection angle (M_PROFILE_PROJECTION_ANGLE). That is, *LowerLimit* = M_PROFILE_PROJECTION_ANGLE - M_PROFILE_PROJECTION_ANGLE_DELTA_NEG. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 170.0	Specifies the lower limit of the projection angle range, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE_DELTA_POS +		Inquires the upper limit of the projection angle range, relative to the nominal projection angle (M_PROFILE_PROJECTION_ANGLE). That is, *UpperLimit* = M_PROFILE_PROJECTION_ANGLE + M_PROFILE_PROJECTION_ANGLE_DELTA_POS. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 170.0	Specifies the upper limit of the projection angle range, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE_MODE +		Inquires whether to consider projection angles within an angular range when extracting the profile to search. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to use multiple projection angles.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to use multiple projection angles.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_PROJECTION_ANGLE_TOLERANCE +		Inquires the tolerance used to determine the projection angles. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* <= 170.0	Specifies the tolerance, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_SAMPLE_SIZE +		Inquires the sampling distance between two consecutive points along the profile line. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the distance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_SAMPLE_SIZE_MODE +		Inquires how to interpret the specified sampling distance (M_PROFILE_SAMPLE_SIZE). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ABSOLUTE	Specifies to use the sampling distance directly, without applying a multiplying factor. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the sampling distance by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the sampling distance by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the sampling distance by the pixel size in X. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the sampling distance by the pixel size in Y. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_THICKNESS +		Inquires the thickness of the profile line. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the thickness. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_PROFILE_THICKNESS_MODE +		Inquires how to interpret the specified thickness value (M_PROFILE_THICKNESS). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ABSOLUTE	Specifies to use the thickness value directly, without applying a multiplying factor. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the thickness value by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b

	M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the thickness value by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the thickness value by the pixel size in X. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the thickness value by the pixel size in Y. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_SPACING	Specifies to multiply the thickness value by the profile spacing. INFO	c		e	f	g	a		i		j	k	l	m	n	b

For a template in a template 3D measurement context, the InquireType parameter can be set to one of the following. PathOrTemplateIndex must be set to the index of a template, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a template in a template 3D measurement context																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_TEMPLATE_EXTENDED_SEARCH +	Inquires whether to extend the search outside of the template's defined area. SET  															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to extend the search beyond the template.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to extend the search beyond the template.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_COUNT +	Inquires the number of profiles used to find the markers of a template.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_UNKNOWN	Specifies that the number of profiles used is unknown. This occurs when the number of profiles is dependent on the depth map calibration. For example, when M_TEMPLATE_PROFILE_NUMBER_MODE is set to M_SPACING and M_TEMPLATE_PROFILE_SPACING_MODE is set to M_RELATIVE_TO_PIXEL_SIZE_....	c		e	f	g	a		i		j	k	l	m	n	b

Value > 0	Specifies the number profiles used.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_NUMBER_ELEMENTS +	Inquires the number of profile elements in the template 3D measurement context. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= M_TEMPLATE_PROFILE_NUMBER_VALUE	Specifies the number of profile elements.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_NUMBER_MODE +	Inquires the mode with which to determine the number of profiles to use to find the markers of a template. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_SPACING	Specifies that the number of profiles is determined based on the spacing between the profiles.	c		e	f	g	a		i		j	k	l	m	n	b
M_USER_DEFINED	Specifies to use the value set with M_TEMPLATE_PROFILE_NUMBER_VALUE. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_NUMBER_VALUE +	Inquires the number of profiles perpendicular to the template. Note that this inquire type is only valid if M_TEMPLATE_PROFILE_NUMBER_MODE is set to M_USER_DEFINED. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
0 < *Value* <= M_TEMPLATE_PROFILE_NUMBER_ELEMENTS	Specifies the number of profiles.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_SPACING +	Inquires the spacing between the profiles along the template. Note that this inquire type is only valid if M_TEMPLATE_PROFILE_NUMBER_MODE is set to M_SPACING. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the spacing. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_SPACING_MODE +	Inquires how to interpret the specified spacing value (M_TEMPLATE_PROFILE_SPACING). Note that this inquire type is only valid if M_TEMPLATE_PROFILE_NUMBER_MODE is set to M_SPACING. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_ABSOLUTE	Specifies to use the spacing value directly, without applying a multiplying factor. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the spacing value by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the spacing value by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the spacing value by the pixel size in X. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the spacing value by the pixel size in Y. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_THICKNESS	Specifies to multiply the spacing value by the profile thickness. INFO	c		e	f	g	a		i		j	k	l	m	n	b

For a template or profile in a template 3D measurement context, the InquireType parameter can be set to one of the following. PathOrTemplateIndex must be set to the index of a template. If ProfileIndex is set to M_DEFAULT, these inquire types allow you to retrieve global settings for the template's perpendicular profiles. If ProfileIndex is set to a specific index, they allow you to retrieve custom settings for the specified profile. Note that any custom setting for the specified profile will only be used if its corresponding M_TEMPLATE_PROFILE_..._SOURCE control is set to M_PROFILE.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a template or profile in a template 3D measurement context																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_TEMPLATE_PROFILE_LENGTH +		Inquires the nominal length of the profile to extract (perpendicular to the template). SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the length. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_LENGTH_MARGIN +		Inquires the margin on the profile length. The nominal length (M_TEMPLATE_PROFILE_LENGTH), plus the margin define the length of the profile. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the margin, as a percentage of the nominal length.	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_LENGTH_MODE +		Inquires how to interpret the specified length value (M_TEMPLATE_PROFILE_LENGTH). SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_ABSOLUTE	Specifies to use the length value directly, without applying a multiplying factor. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the length value by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the length value by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the length value by the pixel size in X. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the length value by the pixel size in Y. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_POSITION_RELATIVE +		Inquires the position of the profile, relative to the template. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	-50.0 <= *Value* <= 50.0	Specifies the relative position, as a percentage of the nominal length.	c		e	f	g	a		i		j	k	l	m	n	b

For a profile perpendicular to a template in a template 3D measurement context, the InquireType parameter can be set to one of the following. PathOrTemplateIndex must be set to the index of a template, and ProfileIndex must be set to the index of one or all profiles. These inquire types allow you to retrieve whether the global or custom settings will be used for the specified profile, when extracting the profile perpendicular to the template.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a profile in a template 3D measurement context																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_TEMPLATE_PROFILE_LENGTH_MARGIN_SOURCE +		Inquires whether to use the global margin value or a custom margin for the profile. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_PROFILE	Specifies to use the custom margin value for the specified profile. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_TEMPLATE	Specifies to use the template's global margin value for the specified profile. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_LENGTH_SOURCE +		Inquires whether to use the global nominal length value and length mode or a custom nominal length and length mode for the profile. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_PROFILE	Specifies to use the custom profile length for the specified profile. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_TEMPLATE	Specifies to use the template's global profile length for the specified profile. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_TEMPLATE_PROFILE_POSITION_SOURCE +		Inquires whether to use the global relative position value or a custom relative position for the profile. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_PROFILE	Specifies to use the custom relative position value for the specified profile. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_TEMPLATE	Specifies to use the template's global relative position value. INFO	c		e	f	g	a		i		j	k	l	m	n	b

For a fit 3D measurement context, the InquireType parameter can be set to one of the following. PathOrTemplateIndex must be set to M_DEFAULT or M_CONTEXT, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a fit 3D measurement context																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_EXPECTED_OUTLIER_PERCENTAGE +		Inquires the expected percentage of outlier markers among the markers that are extracted. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 < *Value* < 100.0	Specifies the expected outliers, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_COVERAGE_MIN +		Inquires the minimum percentage of profiles that must have a marker covered by a fitted point. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the minimum coverage, as a percentage.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE +		Inquires the distance value that Aurora Imaging Library uses to establish outlier markers, which are excluded from the robust best fit. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies that the fit distance is infinite. INFO	c		e	f	g	a		i		j	k	l	m	n	b

Value > 0.0	Specifies the fit distance.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIT_DISTANCE_MODE +	Inquires how to interpret the specified fit distance value (M_FIT_DISTANCE). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_ABSOLUTE	Specifies to use the fit distance value directly, without applying a multiplying factor. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_AUTO	Specifies to automatically calculate the fit distance; the distance is estimated from all the extracted markers. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_MAX	Specifies to multiply the fit distance value by the pixel size in X or in Y, depending on which is bigger.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_MIN	Specifies to multiply the fit distance value by the pixel size in X or in Y, depending on which is smaller.	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_X	Specifies to multiply the fit distance value by the pixel size in X. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_PIXEL_SIZE_Y	Specifies to multiply the fit distance value by the pixel size in Y. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_RELATIVE_TO_SAMPLING	Specifies to multiply the fit distance value by the sampling distance along each profile. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_REMOVE_OUTLIERS +	Inquires whether to remove outlier markers from the template 3D measurement result buffer after the fit. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to remove outliers.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to remove outliers.	c		e	f	g	a		i		j	k	l	m	n	b

For a path 3D measurement context, the InquireType parameter can be set to one of the following. PathOrTemplateIndex must be set to M_DEFAULT or M_CONTEXT, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a path 3D measurement context																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_NUMBER_OF_PATHS +	Inquires the number of paths in the path 3D measurement context.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	Value >= 0	Specifies the number of paths.	c		e	f	g	a		i		j	k	l	m	n	b

For a template 3D measurement context, the InquireType parameter can be set to one of the following. PathOrTemplateIndex must be set to M_DEFAULT or M_CONTEXT, and ProfileIndex must be set to M_DEFAULT.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about a template 3D measurement context																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_NUMBER_OF_TEMPLATES +		Inquires the number of templates in the template 3D measurement context.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0	Specifies the number of templates.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring about a global setting of a profile, path, or template 3D measurement context
- For inquiring about the default template in a profile 3D measurement context, path in a path 3D measurement context, or template in a template 3D measurement context
- For inquiring about the profile extraction along a path in a path 3D measurement context or perpendicular to the template in a template 3D measurement context
- For inquiring about a template in a template 3D measurement context
- For inquiring about a template or profile in a template 3D measurement context
- For inquiring about a profile in a template 3D measurement context
- For inquiring about a fit 3D measurement context
- For inquiring about a path 3D measurement context
- For inquiring about a template 3D measurement context

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to get information about the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the default value																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_DEFAULT		Inquires the default value of the specified inquire type.															
	UserVarPtr extra info	Data type info: Data type: *defined elsewhere*	c		e	f	g	a		i		j	k	l	m	n	b
M_IS_SET_TO_DEFAULT		Inquires whether the specified inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRUE	Specifies that the inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring about a global setting of a profile, path, or template 3D measurement context
- For inquiring about the default template in a profile 3D measurement context, path in a path 3D measurement context, or template in a template 3D measurement context
- For inquiring about the profile extraction along a path in a path 3D measurement context or perpendicular to the template in a template 3D measurement context
- For inquiring about a template in a template 3D measurement context
- For inquiring about a template or profile in a template 3D measurement context
- For inquiring about a profile in a template 3D measurement context
- For inquiring about a fit 3D measurement context
- For inquiring about a path 3D measurement context
- For inquiring about a template 3D measurement context

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to determine whether an inquire type is supported.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To inquire whether an inquire type is supported																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_HAS_DEFAULT		Inquires whether the specified inquire type has a default value.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type does not have a default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type has a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_SUPPORTED		Inquires whether the specified inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring about a global setting of a profile, path, or template 3D measurement context
- For inquiring about the default template in a profile 3D measurement context, path in a path 3D measurement context, or template in a template 3D measurement context
- For inquiring about the profile extraction along a path in a path 3D measurement context or perpendicular to the template in a template 3D measurement context
- For inquiring about a template in a template 3D measurement context
- For inquiring about a template or profile in a template 3D measurement context
- For inquiring about a profile in a template 3D measurement context
- For inquiring about a fit 3D measurement context
- For inquiring about a path 3D measurement context
- For inquiring about a template 3D measurement context

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested information to a required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

To specify the data type																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_FLOAT	Casts the requested information to an *AIL_FLOAT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_FLOAT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT64*. If the requested information does not fit into an *AIL_INT64*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasInquireDraw

Synopsis

Inquire about a setting of a draw 3D measurement context.

Syntax

```
AIL_INT64 M3dmeasInquireDraw(
    AIL_ID DrawContext3dmeasId, //in
    AIL_INT64 Operation,        //in
    AIL_INT64 InquireType,      //in
    void *UserVarPtr            //out
)
```

Description

This function inquires about a specified setting of a draw 3D measurement context. These settings establish which 3D measurement results to draw into the 3D graphics list when calling M3dmeasDraw3d() and how to draw them.

You can control the draw 3D measurement context settings using M3dmeasControlDraw().

Parameters

DrawContext3dmeasId

Specifies the identifier of the draw 3D measurement context to inquire. The draw 3D measurement context must have been previously allocated on the required system using M3dmeasAlloc() with M_DRAW_3D_PROFILE_CONTEXT, M_DRAW_3D_PATH_CONTEXT, or M_DRAW_3D_TEMPLATE_CONTEXT.

Operation

Specifies the draw operation about which to inquire.

See the Parameter associations section for possible values that can be specified.

InquireType

Specifies the type of setting about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_FLOAT
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **M3dmeasInquireDraw**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **Operation** and **InquireType** parameters and possible values returned to the **UserVarPtr** parameter.

- For inquiring a draw profile, path, or template 3D measurement context
- For inquiring a draw path 3D measurement context
- For inquiring a draw template 3D measurement context

The following Operation and InquireType parameter settings are available to inquire about draw profile, path, or template 3D

measurement context settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a draw profile, path, or template 3D measurement context

Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType																
UserVarPtr - Possible values returned																
M_DRAW_MARKERS	Inquires whether and how to draw the cross-like symbol at each specified marker's position, and for pair markers, a rectangle connecting the two transitions with a cross-like symbol at each transition position, when calling M3dmeasDraw3d().															
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b

M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MARKERS_POSITION	Inquires whether and how to draw the cross-like symbol at each specified marker's position when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

	M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MARKERS_WIDTH		Inquires whether and how to draw the width of each specified pair marker when calling M3dmeasDraw3d(). The width is drawn as a rectangle connecting the two transitions with a cross-like symbol at each transition position. This operation is only supported for pair markers.	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +		Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PROFILE_EXTRACTED		Inquires whether and how to draw the extracted points that form each specified profile when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +		Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PROFILE_REGION	Inquires whether and how to draw the region where each specified profile was extracted when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

		M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PROFILE_RESPONSE	Inquires whether and how to draw the response of each specified profile (after applying the filter) when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_APPEARANCE +		Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_TRANSITIONS		Inquires whether and how to draw the step-like symbol at each specified marker's transition position when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

		M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_APPEARANCE +		Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_GLOBAL_DRAW_SETTINGS	Inquires a global draw 3D measurement context setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_COLOR_OFFSET +	Inquires the index of the volume element to draw. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0	Specifies the offset to apply to the index.	c		e	f	g	a		i		j	k	l	m	n	b

The following Operation and InquireType parameter settings are available to inquire about draw path 3D measurement context settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a draw path 3D measurement context																	
Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType																	
UserVarPtr - Possible values returned																	
M_DRAW_PATH	Inquires whether and how to draw the path defined in the path 3D measurement context when calling M3dmeasDraw3d().																
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b

	M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_PATH_PROJECTED		Inquires whether and how to draw the projection of the path defined in the path 3D measurement context when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +		Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b

The following Operation and InquireType parameter settings are available to inquire about draw template 3D measurement context settings.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring a draw template 3D measurement context																
Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
InquireType																
UserVarPtr - Possible values returned																
M_DRAW_FIND_GEOMETRY	Inquires whether and how to draw the line segments connecting the extracted markers when calling M3dmeasDraw3d().															
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b

		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIND_GEOMETRY_PROJECTED			Inquires whether and how to draw the planes connecting the projections of the extracted markers when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +		Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_CYAN	Specifies the color dark cyan.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_FIT_GEOMETRY	Inquires whether and how to draw the fitted 3D geometry when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIT_GEOMETRY_PROJECTED	Inquires whether and how to draw the projection of the fitted 3D geometry when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b

M_OPACITY +	Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +	Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIT_MARKERS	Inquires whether and how to draw the cross-like symbol at the position of each marker that was used for the fit operation when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b

	M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c	e	f	g	a	i	j	k	l	m	n	b
	M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c	e	f	g	a	i	j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c	e	f	g	a	i	j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c	e	f	g	a	i	j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c	e	f	g	a	i	j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c	e	f	g	a	i	j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c	e	f	g	a	i	j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c	e	f	g	a	i	j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c	e	f	g	a	i	j	k	l	m	n	b
	Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c	e	f	g	a	i	j	k	l	m	n	b
M_DRAW_FIT_MARKERS_OUTLIERS		Inquires whether and how to draw the cross-like symbol at the position of each marker that was considered an outlier for the fit operation when calling M3dmeasDraw3d().	c	e	f	g	a	i	j	k	l	m	n	b
M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c	e	f	g	a	i	j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	e	f	g	a	i	j	k	l	m	n	b
	M_DISABLE	Specifies not to draw the graphic(s). INFO	c	e	f	g	a	i	j	k	l	m	n	b
	M_ENABLE	Specifies to draw the graphic(s).	c	e	f	g	a	i	j	k	l	m	n	b
M_APPEARANCE +		Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c	e	f	g	a	i	j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c	e	f	g	a	i	j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c	e	f	g	a	i	j	k	l	m	n	b
	M_POINTS	Specifies a points appearance. INFO	c	e	f	g	a	i	j	k	l	m	n	b

		M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR +		Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_RED	Specifies the color red.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_TEMPLATE			Inquires whether and how to draw the template defined in the template 3D measurement context when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACTIVE +		Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
		M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ENABLE	Specifies to draw the graphic(s).	c		e	f	g	a		i		j	k	l	m	n	b

M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
	M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
		UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_TEMPLATE_PROJECTED			Inquires whether and how to draw the projection of the template defined in the template 3D measurement context when calling M3dmeasDraw3d().	c		e	f	g	a		i		j	k	l	m	n	b

M_ACTIVE +	Inquires whether to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies not to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies to draw the graphic(s). INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_APPEARANCE +	Inquires whether to draw the graphic(s) as a solid surface, wireframe, or points. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
M_POINTS	Specifies a points appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID	Specifies a solid appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_POINTS	Specifies a solid appearance with points. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SOLID_WITH_WIREFRAME	Specifies a solid appearance with a wireframe. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_WIREFRAME	Specifies a wireframe appearance. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR +	Inquires the color with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLACK	Specifies the color black. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BLUE	Specifies the color blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_BRIGHT_GRAY	Specifies the color bright gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_CYAN	Specifies the color cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_BLUE	Specifies the color dark blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_CYAN	Specifies the color dark cyan. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_GREEN	Specifies the color dark green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_MAGENTA	Specifies the color dark magenta. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_COLOR_DARK_RED	Specifies the color dark red. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_COLOR_DARK_YELLOW	Specifies the color dark yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GRAY	Specifies the color gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_GREEN	Specifies the color green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_BLUE	Specifies the color light blue. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GRAY	Specifies the color light gray. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_GREEN	Specifies the color light green. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_LIGHT_WHITE	Specifies the color light white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_MAGENTA	Specifies the color magenta.	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_RED	Specifies the color red. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_WHITE	Specifies the color white. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_COLOR_YELLOW	Specifies the color yellow. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different color, according to the mapping between each result element's index and a distinct color in the M_COLORMAP_DISTINCT_256 colormap.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PSEUDO_DARK_COLOR	Specifies to draw the graphic for each 3D measurement result element with a different dark color, according to the mapping between each element's index and a distinct dark color corresponding to those in the M_COLORMAP_DISTINCT_256 colormap, where the dark color has the same hue but a lower luminance.	c		e	f	g	a		i		j	k	l	m	n	b
M_OPACITY +		Inquires the opacity with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the opacity of the 3D graphic. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THICKNESS +		Inquires the thickness with which to draw the graphic(s). SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

		M_GRAPHIC_LIST_DEFAULT	Specifies to use the 3D graphics list's default setting.	c		e	f	g	a		i		j	k	l	m	n	b
		Value > 0.0	Specifies the point/line thickness, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring a draw profile, path, or template 3D measurement context
- For inquiring a draw path 3D measurement context
- For inquiring a draw template 3D measurement context

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to get information about the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the default value																		
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																	
M_DEFAULT		Inquires the default value of the specified inquire type.																
	UserVarPtr extra info	Data type info: Data type: *defined elsewhere*	c		e	f	g	a		i		j	k	l	m	n	b
M_IS_SET_TO_DEFAULT		Inquires whether the specified inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is set to its default value.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring a draw profile, path, or template 3D measurement context
- For inquiring a draw path 3D measurement context
- For inquiring a draw template 3D measurement context

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to determine whether an inquire type has a default value or whether it is supported.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring whether an inquire type has a default value or whether it is supported																		
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																	
M_HAS_DEFAULT		Inquires whether the specified inquire type has a default value.																

UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the inquire type does not have a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the inquire type has a default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_SUPPORTED	Inquires whether the specified inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For inquiring a draw profile, path, or template 3D measurement context
- For inquiring a draw path 3D measurement context
- For inquiring a draw template 3D measurement context

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested information to the required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_FLOAT	Casts the requested information to an *AIL_FLOAT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_FLOAT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT64*. If the requested information does not fit into an *AIL_INT64*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasRestore

Synopsis

Restore a 3D measurement context from disk.

Syntax

```
AIL_ID M3dmeasRestore(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID SysId,                //in
    AIL_INT64 ControlFlag,       //in
    AIL_ID *Context3dmeasIdPtr   //out
)
```

Description

This function restores a 3D measurement context that was previously saved to a file, using M3dmeasSave() or M3dmeasStream(). This function restores all of the 3D measurement context's settings that were in effect when it was saved.

When the restored 3D measurement context is no longer required, release it using M3dmeasFree() unless M_UNIQUE_ID was specified during restoration; if M_UNIQUE_ID was specified, the smart identifier manages the 3D measurement context's lifetime and you must not manually free it.

Parameters

FileName

Specifies the name and path of the file from which to restore the 3D measurement context. The function handles (internally) the opening and closing of the file.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Open** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") [1]	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). Typically, 3D measurement files have an M3DMEAS extension. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

SysId

Specifies the system on which to restore the 3D measurement context.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system identifier																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Context3dmeasIdPtr

Specifies the address of the variable in which to write the 3D measurement context identifier or specifies the data type that the function should use to return the 3D measurement context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D measurement context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated 3D measurement context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_3DMEAS_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the 3D measurement context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the draw 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated draw 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the find marker 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated find marker 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Address in which to write the fit 3D measurement context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated fit 3D measurement context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the 3D measurement context identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_3DMEAS_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasSave

Synopsis

Save a 3D measurement context to a file.

Syntax

```
void M3dmeasSave(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID Context3dmeasId,      //in
    AIL_INT64 ControlFlag        //in
)
```

Description

This function saves all the information about a previously allocated 3D measurement context to disk.

To load a saved context, use either M3dmeasRestore() or M3dmeasStream().

Parameters

FileName

Specifies the name and path of the file in which to save the 3D measurement context; it is recommended that you use the M3DMEAS file extension for easier use with other Aurora Imaging software products. The function internally handles the opening and closing of this file. If this file already exists, it will be overwritten.

This parameter can be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Save As** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). The recommended extension is M3DMEAS. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Context3dmeasId

Specifies the identifier of the 3D measurement context to save. The context must have been successfully allocated with M3dmeasAlloc() prior to calling this function.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.

DLL	Requires ail11.dll; ail3dmeas11.dll.

M3dmeasStream

Synopsis

Load, restore, or save a 3D measurement context from/to a file or memory stream.

Syntax

```
void M3dmeasStream(
    AIL_TEXT_PTR MemPtrOrFileName, //in-out
    AIL_ID SysId,                  //in
    AIL_INT64 Operation,           //in
    AIL_INT64 StreamType,          //in
    AIL_DOUBLE Version,            //in
    AIL_INT64 ControlFlag,         //in
    AIL_ID *Context3dmeasIdPtr,    //in-out
    AIL_INT *SizeByteVarPtr        //out
)
```

Description

This function can load, restore, or save a 3D measurement context from/to a file or memory stream.

To inquire the number of bytes necessary to save a 3D measurement context to a memory stream, you should first call this function (**M3dmeasStream**()) with M_INQUIRE_SIZE_BYTE.

The content saved to memory stream is equivalent to the content saved to file. This function is equivalent to a file saved using M3dmeasSave().

You can use this and other Aurora Imaging Library stream functions, for example, to save all required Aurora Imaging Library objects, as well as any other custom data, for your application to a memory stream. Once in a memory stream, you can write the stream to a single file or transfer it over a network. You are responsible for concatenating the streams and for saving the stream to file.

Using **M3dmeasStream**(), you can choose to save a backwards-compatible version of the 3D measurement context, which will work using a version of Aurora Imaging Library that is up to one major release older than the current version (depending on which version is specified). However, all settings and features unique to the higher version will be ignored when restored using the lower version. Besides saving backwards-compatible versions, you can also load or restore 3D measurement contexts saved using any Aurora Imaging Library version with which this function was available. Settings that do not exist in the lower version will be filled with default values when the 3D measurement context is loaded or restored.

Parameters

MemPtrOrFileName
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_TEXT_PTR [optionally, in C++: a reference to an AIL_STRING]
- Optionally, in C++: a reference to a std::vector of type AIL_UINT8

Specifies the file or memory stream.

See the Parameter associations section for possible values that can be returned/specified.

SysId

Specifies the system on which to restore the 3D measurement context.

See the Parameter associations section for possible values that can be specified.

Operation

Specifies the operation to perform.

See the Parameter associations section for possible values that can be specified.

StreamType

Specifies the type of stream in which to store/from which to restore the 3D measurement context. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of stream																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILE	Specifies a file stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_MEMORY	Specifies a memory stream. You are responsible for allocating a block of memory for the stream.	c		e	f	g	a		i		j	k	l	m	n	b

Version

Specifies the Aurora Imaging Library version of the 3D measurement context.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Reserved for future expansion. This parameter must be set to **M_DEFAULT**.

Context3dmeasIdPtr

Specifies the address of the variable in which to write or from which to read the identifier of the 3D measurement context. If the M_RESTORE operation is not successful, **M_NULL** is returned.

See the Parameter associations section for possible values that can be returned/specified.

SizeByteVarPtr

Specifies the address of the variable in which to write the size of the 3D measurement context, in bytes.

If the size is not required, you can set this parameter to **M_NULL**.

> Note that the size of the 3D measurement context will vary depending on the Aurora Imaging Library version specified.

Parameter associations

The table below lists possible values for the **MemPtrOrFileName**, **SysId**, **Operation**, **Version**, and **Context3dmeasIdPtr** parameters.

- For performing the stream operation.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For performing the stream operation.																
Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
MemPtrOrFileName - Possible values passed or returned																
SysId																
Version																
Context3dmeasIdPtr - Possible values passed or returned																

M_INQUIRE_SIZE_BYTE	Inquires the number of bytes required to save a 3D measurement context to memory stream. This operation is not supported when the StreamType parameter is set to M_FILE.														
MemPtrOrFileName	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	Specifies the Aurora Imaging Library version with which the 3D measurement context must be compatible when saved. The size of a 3D measurement context might differ from one Aurora Imaging Library version to another.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
Context3dmeasIdPtr	Specifies the address of the variable containing the identifier of the 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOAD	Loads the content of a specified file or memory stream into a previously allocated 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream from which to load the 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. 3D measurement contexts typically have an M3DMEAS file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b

SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
Context3dmeasIdPtr	Specifies the address of the variable containing the identifier of the 3D measurement context. All previous settings are overwritten with those of the loaded 3D measurement context, or are set to the default value if the settings didn't exist in the version of Aurora Imaging Library from which the 3D measurement context is being loaded.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESTORE	Restores a 3D measurement context from a file or memory stream and assigns it an Aurora Imaging Library identifier.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream from which to restore the 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. 3D measurement contexts typically have an M3DMEAS file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	Specifies the system on which to allocate the 3D measurement context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b
Version	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b

Context3dmeasIdPtr	Specifies the address of the variable in which to write the identifier of the 3D measurement context. If the operation is not successful, **M_NULL** is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE	Saves a 3D measurement context to a specified file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. For easier use with other Aurora Imaging software products, when saving a 3D measurement context to a file, use the M3DMEAS file extension. The function handles (internally) the opening and closing of the file. If the file already exists, it will be overwritten. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which to write, and the block of memory must be large enough to stream the entire object. To determine the required size, call this function with M_INQUIRE_SIZE_BYTE. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	Specifies the Aurora Imaging Library version with which the 3D measurement context must be compatible when saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
Context3dmeasIdPtr	Specifies the address of the variable containing the identifier of the 3D measurement context to save.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ail3dmeas11.lib.
DLL	Requires ail11.dll; ail3dmeas11.dll.